# The leximin core selection on exact games

Draft fragment: boundary theorems, exact bump paths, and the open crux

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*Status: mathematical core of a paper in progress. Singleton bumps are proved monotone even on balanced games. Grand-coalition bumps remain on an exact path but labeled resource monotonicity is disproved by an exact five-player example. Proper nonsingleton bumps are reduced to an exact capacity dichotomy and one open path lemma; the capacity implication is proved for every two-level failure dual, including all two-player coalitions, and a sharp quantitative carrier–star inequality is proved for two-player coalitions. The quantitative carrier inequality for arbitrary target coalitions is proved exactly through six players and disproved by exact three-generator games in every dimension from seven onward. Balanced nonexact counterexamples establish that exactness is necessary for the remaining positive program.*

## 1 The leximin-above-floors core selection

Let $(N, v)$ be a balanced TU game, $M = v(N)$, and floors $m_i := v(\{i\})$. For $x \in \mathbb{R}^N$ write the *gains* $g_i(x) = x_i - m_i$. The *leximin-above-floors* selection is

$$\phi(v) := \arg \operatorname*{leximin}_{x \in \mathrm{Core}(v)} \big(g_1(x), \dots, g_n(x)\big),$$

the unique core allocation whose ascending-sorted gain vector is lexicographically largest. It is single-valued (leximin over a polytope) and is computed by the Maschler-type scheme over player coordinates: repeatedly raise a common lower level $s$ of the free players' gains as far as the core allows, fix ("pin") every free player whose gain cannot exceed $m_i + s$ on the level-optimal face, and recurse on the rest. On *exact* games $m_i = \min_{\mathrm{Core}} x_i$, so $g_i \geq 0$ and $\phi$ egalitarianly distributes the surplus $x_i - v(\{i\})$ inside the core; on convex games it is the constrained egalitarian solution of Dutta and Ray (1989). Implementation (exact rational + float) in `src/leximin.py`, validated against those two coincidences.

We ask whether $\phi$ is *coalitionally monotonic*: if $v' = v + d\,\mathbf{1}_S$ for a single coalition $S$ and $d > 0$, is $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$? A positive answer on exact games would exhibit an explicit coalitionally monotonic core selection on that class. The reduction in Section 5 addresses only proper nonsingleton coalitions. Singleton bumps change the floors defining $\phi$, and a grand-coalition bump changes the efficiency hyperplane. The singleton case is resolved next. The grand-coalition case has a negative answer even on a simple exact five-player game (Theorem 3.2).

## 2 Singleton bumps are monotone

We first isolate an elementary property of the leximin order. Write $a \succ_{\mathrm{lx}} b$ when the ascending rearrangement of $a$ is lexicographically larger than that of $b$.

**Lemma 2.1** (one-coordinate translation)**.** *If $a \succ_{\mathrm{lx}} b$, $a_i < b_i$, and $d > 0$, then $a + de_i \succ_{\mathrm{lx}} b + de_i$.*

*Proof.* For a finite vector $z$, put $L_\lambda(z) = \sum_j e^{-\lambda z_j}$. The first unequal coordinate of the ascending rearrangements dominates this sum as $\lambda \to \infty$; hence $a \succ_{\mathrm{lx}} b$ if and only if $L_\lambda(a) < L_\lambda(b)$ for all sufficiently large $\lambda$. Moreover,

$$L_\lambda(b + de_i) - L_\lambda(a + de_i) = L_\lambda(b) - L_\lambda(a) + (1 - e^{-\lambda d})(e^{-\lambda a_i} - e^{-\lambda b_i}).$$

Both terms on the right are positive for all sufficiently large $\lambda$, because $a_i < b_i$. The same characterization gives the claim. □

**Theorem 2.2** (singleton coalitional monotonicity)**.** *Let $v$ and $v' = v + d\,\mathbf{1}_{\{i\}}$ be balanced games, with $d > 0$. Then*

$$\phi_i(v') \geq \phi_i(v).$$

*Thus singleton bumps are monotone even without exactness.*

*Proof.* Let $m_j = v(\{j\})$ and express the old core in gain coordinates

$$G = \{(x_j - m_j)_{j \in N} : x \in \mathrm{Core}(v)\}.$$

If the singleton floor of player $i$ is raised by $d$, the new feasible gain set is

$$H = \{g - de_i : g \in G,\ g_i \geq d\}.$$

Let $z$ be the leximin point of $G$ and $h$ the leximin point of $H$. The desired inequality is $h_i + d \geq z_i$. If $z_i < d$, this is immediate from $h_i \geq 0$. Suppose $z_i \geq d$. Then $b = z - de_i$ belongs to $H$, so $h$ is either $b$ or $h \succ_{\mathrm{lx}} b$. If, contrary to the claim, $h_i + d < z_i$, then $h_i < b_i$ and the equality case is impossible. Lemma 2.1 gives

$$h + de_i \succ_{\mathrm{lx}} b + de_i = z.$$

But $h + de_i \in G$, contradicting the definition of $z$ as the leximin point of $G$. □

## 3 Grand-coalition bumps: an exact path but no resource monotonicity

The grand-coalition case also has an exact path, but its geometry differs from the nested-core path of Section 5.

**Lemma 3.1** (grand-bump exact path)**.** *Let $v$ be exact and, for $r \geq 0$, let $v^r$ agree with $v$ on every proper coalition and satisfy $v^r(N) = v(N) + r$. Then $v^r$ is exact for every $r \geq 0$. Moreover,*

$$\mathrm{Core}(v) + r\Delta_N \subseteq \mathrm{Core}(v^r), \qquad \Delta_N = \{a \in \mathbb{R}^N_+ : a(N) = 1\}.$$

*Proof.* If $x \in \mathrm{Core}(v)$ and $a \in r\Delta_N$, then $x + a$ has total $v(N) + r$ and weakly raises every coalition sum, proving the inclusion and balancedness. Fix a proper coalition $T$. Exactness of $v$ supplies $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$. Choose $j \notin T$. Then $x^T + re_j \in \mathrm{Core}(v^r)$ by the inclusion and remains tight at $T$. Thus

$$\min_{x \in \mathrm{Core}(v^r)} x(T) = v(T) = v^r(T).$$

The grand coalition is tight throughout its core by efficiency, so $v^r$ is exact at every coalition. □

The exact path does not rescue labeled resource monotonicity.

**Theorem 3.2** (grand resource monotonicity fails on exact games)**.** *There is an exact five-player game $v$ such that, when only $v(N)$ is raised, one player's leximin-above-floors payoff strictly falls. Indeed, let*

$$p = (0, 0, 7, 0, 7), \qquad q = (2, 10, 0, 2, 0),$$

*put $v(N) = 14$, and for every proper $T$ put $v(T) = \min\{p(T), q(T)\}$. If $v^r(N) = 14 + r$ and $v^r(T) = v(T)$ for $T \subsetneq N$, then for $0 \le \varepsilon \le 3/2$,*

$$\phi(v^{2+\varepsilon}) = \left(3 + \frac{\varepsilon}{3}, 4 - \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}\right).$$

*Thus player 2 falls at rate $1/3$ while the grand worth rises. For example, raising the grand worth from 16 to $129/8$ changes that player's payoff from 4 to $95/24$.*

*Proof.* Both $p$ and $q$ have total 14 and belong to $\mathrm{Core}(v)$. At every proper coalition one of them attains $v(T)$, so $v$ is exact; every $v^r$ is exact by Lemma 3.1. All singleton worths are zero.

Write $s = 3 + \varepsilon/3$ and $z = (s, 7 - s, s, s, s)$. Direct substitution in the lower-envelope definition of $v$ shows that $z \in \mathrm{Core}(v^{2+\varepsilon})$ throughout the stated interval. (At $\varepsilon = 0$ every core slack is nonnegative; the only coalition sum with negative slope is the singleton of player 2, whose slack remains positive. Every other slope is nonnegative.)

It remains to identify the leximin point. The two pair constraints

$$x_2 + x_3 \ge 7, \qquad x_2 + x_5 \ge 7$$

belong to the core. If every coordinate of a core point is at least $t$, then efficiency and the first constraint give $16 + \varepsilon = x(N) \ge 7 + 3t$, hence $t \le s$. So $z$ attains the largest possible minimum coordinate. On that level-optimal face put $d_i = x_i - s \ge 0$ and $D = 1 - 2\varepsilon/3 \ge 0$. Efficiency gives $\sum_i d_i = D$, while the two displayed constraints give $d_2 + d_3 \ge D$ and $d_2 + d_5 \ge D$. Nonnegativity forces $d_2 = D$ and every other $d_i = 0$. Hence the level-optimal face is the singleton $\{z\}$, proving the formula and the strict drop. □

The counterexample pinpoints the failure mechanism: four players form the lowest leximin stratum, and additional total resource raises them jointly only by taking payoff from a higher-stratum player pinned to two cross-stratum pair constraints. The feasible comparison points supplied by $\mathrm{Core}(v) + r\Delta_N$ therefore do not control labeled coordinates.

## 4 Exactness is necessary

Coalitional monotonicity of $\phi$ fails on balanced games in general—already at $n = 4$, and even at $n = 5$ where every floor is core-attained. So exactness cannot be weakened to balancedness.

**Theorem 4.1.** *There is a balanced (non-exact) five-player game $v$, a coalition $S$ with $|S| = 3$, and $d > 0$ with $v + d\,\mathbf{1}_S$ balanced, such that some $i \in S$ has $\phi_i(v + d\,\mathbf{1}_S) < \phi_i(v)$. The same holds for a four-player game.*

**Example 4.2** ($n = 5$; floors attained)**.** *Let $N = \{1, \dots, 5\}$ with $v(N) = 7$ and (all other coalitions as listed in the supplement) in particular*

$$v(\{1,2,3\}) = 3, \quad v(\{2,3\}) = 3, \quad v(\{1,3,5\}) = v(\{3,5\}) = 4, \quad v(\{1,5\}) = -1.$$

*Floors are $m = (0, 1, 1, 1, 2)$, and $m_i = \min_{\mathrm{Core}} x_i$ for* every *$i$ (all floors attained); the game is balanced but not exact (e.g. $\min_{\mathrm{Core}} x(\{1,5\}) = 1 > -1$). Then*

$$\phi(v) = \left(\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}, 1, \tfrac{5}{2}\right), \qquad \phi(v + \mathbf{1}_{\{1,2,3\}}) = (1, 1, 2, 1, 2),$$

*both in their respective cores, and player $2 \in \{1,2,3\}$* drops *from $\frac{3}{2}$ to $1$ when $v(\{1,2,3\})$ is raised from $3$ to $4$. Thus exactness at the singletons (floor attainment) does not suffice: it is exactness at higher coalitions that the positive theorem needs.*

**Example 4.3** ($n = 4$; floor unattained)**.** *Let $N = \{1,2,3,4\}$, $v(N) = 20$, with $v(\{1\}) = -3$, $v(\{1,2\}) = 11$, $v(\{1,3\}) = 7$, $v(\{2,3\}) = 4$, $v(\{1,2,3\}) = 13$, $v(\{2,4\}) = 10$, $v(\{1,3,4\}) = 9$, $v(\{2,3,4\}) = 13$, and the remaining values in the supplement. Here $v(\{1\}) = -3 < 1 = \min_{\mathrm{Core}} x_1$. Then $\phi(v) = (\frac{5}{2}, \frac{17}{2}, \frac{9}{2}, \frac{9}{2})$ and $\phi(v + 3\,\mathbf{1}_{\{1,2,3\}}) = (2, 9, 5, 4)$: player $1 \in \{1,2,3\}$ drops from $\frac{5}{2}$ to $2$.*

Both examples are certified in exact rational arithmetic ($\phi$ in the core on both sides, the drop exact); see the supplement and `src/hunt_leximin_nonexact.py`.

## 5 Reduction for proper nonsingleton bumps on exact games

Fix an exact $v$, a proper coalition $S$ with $|S| \geq 2$, and $d > 0$ with $v' = v + d\,\mathbf{1}_S$ exact. Since $|S| \geq 2$ the floors are unchanged, and

$$\mathrm{Core}(v') = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\} \subseteq \mathrm{Core}(v).$$

For $\tau \in [v(S), v(S)+d]$ let $v_\tau$ agree with $v$ except $v_\tau(S) = \tau$, so $\mathrm{Core}(v_\tau) = \mathrm{Core}(v) \cap \{x(S) \geq \tau\}$ interpolates $\mathrm{Core}(v)$ and $\mathrm{Core}(v')$.

**Lemma 5.1** (exact interval)**.** *Every $v_\tau$, $\tau \in [v(S), v(S) + d]$, is exact.*

*Proof.* $\mathrm{Core}(v_\tau) \supseteq \mathrm{Core}(v') \neq \emptyset$, so $v_\tau$ is balanced. Exactness needs $\min_{\mathrm{Core}(v_\tau)} x(T) = v_\tau(T)$ for all $T$. For $T = S$, exactness of $v$ supplies $x^0 \in \mathrm{Core}(v)$ with $x^0(S) = v(S)$, while any $x^1 \in \mathrm{Core}(v')$ satisfies $x^1(S) \geq v(S) + d$. The core of $v$ is convex, so the segment $[x^0, x^1]$ contains a point $x^\tau \in \mathrm{Core}(v)$ with $x^\tau(S) = \tau$. This point belongs to $\mathrm{Core}(v_\tau)$, proving $\min_{\mathrm{Core}(v_\tau)} x(S) = \tau$. For $T \neq S$: put $f_T(\tau) = \min_{\mathrm{Core}(v_\tau)} x(T)$. As $\tau$ grows $\mathrm{Core}(v_\tau)$ shrinks, so $f_T$ is nondecreasing; and $\mathrm{Core}(v_\tau) \subseteq \mathrm{Core}(v)$ gives $f_T(\tau) \geq \min_{\mathrm{Core}(v)} x(T) = v(T)$. At the endpoints $f_T = v(T)$ because $v$ and $v'$ are exact. A nondecreasing function equal to $v(T)$ at both endpoints and $\geq v(T)$ between is $\equiv v(T)$. Hence $f_T(\tau) = v(T) = v_\tau(T)$. □

The entire exact-bump range can be read from exposed core faces. For $T \subseteq N$, write $F_T = \{x \in \mathrm{Core}(v) : x(T) = v(T)\}$ (with $F_N = \mathrm{Core}(v)$), and define

$$d_v^*(S) = \min_{T \neq S} \left( \max_{x \in F_T} x(S) - v(S) \right).$$

**Proposition 5.2** (exact bump capacity)**.** *For an exact game $v$ and a proper coalition $S$, the bumped game $v + d\mathbf{1}_S$ is exact if and only if $0 \leq d \leq d_v^*(S)$. If $\sigma_S = \phi(v)(S) - v(S)$, then every exact bump at $S$ is non-binding if and only if $d_v^*(S) \leq \sigma_S$; a binding exact bump exists if and only if $d_v^*(S) > \sigma_S$.*

*Proof.* The bumped core is $K_d = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\}$. For $T \neq S$, exactness at $T$ is equivalent to $F_T \cap K_d \neq \emptyset$, which is equivalent to

$$d \leq \max_{x \in F_T} x(S) - v(S).$$

Taking the minimum over $T \neq S$ gives all the higher-coalition exactness conditions; the term $T = N$ also gives nonemptiness of $K_d$. Exactness at $S$ follows by interpolating between a point of $F_S$ and a point of $K_d$, as in Lemma 5.1. This proves the first statement. The last two statements compare the exact capacity with the largest bump still satisfied by the old selection, namely $\sigma_S$. □

The capacity implication can be proved for every star-failure dual with at most two residual levels. This includes every two-player coalition and also explains the same-stratum pair certificates found in the computation.

**Proposition 5.3** (two-level dual certificate)**.** *Let $v$ be exact, $S$ proper, $\phi = \phi(v)$, and $\sigma_S = \phi(S) - v(S)$. Suppose $(\star_S)$ fails and its dual maximal face can be represented by weights taking at most two values on $S$: after normalization, for some $\emptyset \neq C \subsetneq S$ and $R > 1$,*

$$\phi \in \arg\max_{x \in \mathrm{Core}(v)} \left( x(C) + Rx(S \setminus C) \right).$$

*Then*

$$\max_{x \in F_C} x(S) \leq \phi(S) \quad \textit{and hence} \quad d_v^*(S) \leq \sigma_S.$$

*The same conclusion holds in the one-level case where $\phi$ maximizes $x(S)$.*

*Proof.* Put $H = S \setminus C$ and $h = \phi(C) - v(C) \geq 0$. Exactness makes $F_C = \{x \in \mathrm{Core}(v) : x(C) = v(C)\}$ nonempty. For $x \in F_C$, maximality of $\phi$ gives

$$(x(C) - \phi(C)) + R(x(H) - \phi(H)) \leq 0.$$

The first parenthesis is $-h$, so $x(H) - \phi(H) \leq h/R$. Therefore

$$x(S) - \phi(S) = -h + x(H) - \phi(H) \leq -h \left(1 - \frac{1}{R}\right) \leq 0.$$

Thus $F_C$ certifies the required bound:

$$d_v^*(S) \leq \phi(S) - v(S) = \sigma_S.$$

In the one-level case, $\phi(S) = \max_{\mathrm{Core}(v)} x(S)$, so the conclusion is immediate. □

**Corollary 5.4** (the capacity implication for pairs)**.** *If $v$ is exact and $S$ is a proper two-player coalition, then*

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

*Proof.* When $(\star_S)$ fails, its positive dual weights on the two members of $S$ have either one value or two. Normalize the smaller value to one and apply Proposition 5.3. □

There is a second exact certificate, independent of the number of residual levels. It compares the slack of a coalition carrying the whole complement of $S$ with the slack of its trace inside $S$.

**Proposition 5.5** (off-side slack-domination certificate)**.** *Let $v$ be exact, $S$ proper, $\phi = \phi(v)$, and $P = N \setminus S$. Suppose $C$ is a nonempty proper subset of $S$, put $T = P \cup C$, and assume*

$$\phi(T) - v(T) \le \phi(C) - v(C).$$

*Then*

$$\max_{x \in F_C} x(S) \le \phi(S), \qquad \text{so} \qquad d_v^*(S) \le \phi(S) - v(S).$$

*In particular, the conclusion holds whenever $T$ is $\phi$-tight.*

*Proof.* Because $S \cup T = N$ and $S \cap T = C$, modularity gives $x(S) + x(T) = x(N) + x(C)$ for every allocation $x$. If $x \in F_C$, then core feasibility at $T$ and efficiency imply

$$x(S) = v(N) + v(C) - x(T) \le v(N) + v(C) - v(T).$$

Moreover,

$$v(N) + v(C) - v(T) = \phi(S) + \big(\phi(T) - v(T)\big) - \big(\phi(C) - v(C)\big) \le \phi(S).$$

Exactness makes $F_C$ nonempty, so this face occurs in the minimum defining $d_v^*(S)$ and proves the claim. □

The carrier comparison has a useful normal form that removes both slack expressions. It also converts failure of every candidate into a single finite-dimensional separation problem.

**Proposition 5.6** (carrier marginal and face separation)**.** *Let $v$ be exact, $\phi = \phi(v)$, $S$ proper, and $P = N \setminus S$. For every $k \in S$,*

$$\begin{aligned}[\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] \\ = \phi(P) - [v(P \cup \{k\}) - v(\{k\})].\end{aligned}$$

*Consequently $k$ satisfies the carrier inequality if and only if*

$$v(P \cup \{k\}) - v(\{k\}) \ge \phi(P). \tag{M}$$

*Let $K \subseteq S$ be nonempty. If (M) fails strictly for every $k \in K$, then every point of the exact face $F_{P \cup \{k\}}$ satisfies $x(S) > \phi(S)$. If, in addition, $(\star_S)$ fails, there is a nonzero $w \in \mathbb{R}_+^S$ such that*

$$\max_{k \in K} \max_{x \in F_{P \cup \{k\}}} \sum_{i \in S} w_i (x_i - \phi_i) < 0. \tag{Sep}$$

*Proof.* Because $P$ and $\{k\}$ are disjoint, $\phi(P \cup \{k\}) = \phi(P) + \phi_k$, which proves the identity and (M). Now suppose the latter fails for $k$ and take $x \in F_{P\cup\{k\}}$. Core feasibility at $\{k\}$ gives

$$x(P) = v(P \cup \{k\}) - x_k \leq v(P \cup \{k\}) - v(\{k\}) < \phi(P).$$

Efficiency therefore gives $x(S) > \phi(S)$.

Let $Z$ be the convex hull of the union of the faces $F_{P\cup\{k\}}$, $k \in K$. It is a compact subset of the core, and every point of $Z$ has $S$-sum strictly larger than $\phi(S)$. Hence failure of $(\star_S)$ implies that $Z$ is disjoint from $\phi_S + \mathbb{R}^S_+$ after projection onto the $S$-coordinates. Strong separation gives a strict separating normal $w$. Its coordinates must be nonnegative, since otherwise its infimum on $\phi_S + \mathbb{R}^S_+$ would be $-\infty$. Translating by $\phi_S$ gives (Sep). □

The separator is in fact forced by the star-failure cover; it is not an independent normal that can be eliminated against that cover.

**Proposition 5.7** (residual-gap separator)**.** *Suppose failure of $(\star_S)$ has a tight-coalition certificate*

$$r^S = \mu \mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T, \qquad \lambda_T \geq 0, \quad \phi(T) = v(T) \text{ if } \lambda_T > 0,$$

*where $r_i \geq 1$ on $S$. Put*

$$r_0 = \min_{i\in S} r_i, \qquad K = \{i \in S : r_i = r_0\}, \qquad H = S \setminus K, \qquad D = \sum_{i\in H}(r_i - r_0).$$

*Assume $H \neq \emptyset$. If the carrier marginal fails at $k \in K$ by*

$$\gamma_k = [\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] > 0,$$

*then, for every $x \in F_{P\cup\{k\}}$,*

$$\sum_{i\in H} \frac{r_i - r_0}{D}(x_i - \phi_i) \leq -\frac{r_0}{D}\gamma_k < 0. \tag{RG}$$

*Consequently the normalized residual gap is a common strict separator for every failed candidate in $K$, supported entirely on the higher-residual class $H$.*

*Proof.* For any $x \in \mathrm{Core}(v)$ put $d = x - \phi$. Efficiency gives $d(N) = 0$, and core feasibility plus tightness gives $d(T) \geq 0$ whenever $\lambda_T > 0$. Taking the scalar product of the cover identity with $d$ therefore yields

$$\sum_{i\in S} r_i d_i = -\sum_T \lambda_T d(T) \leq 0.$$

Splitting off the minimum residual gives

$$r_0 d(S) + \sum_{i\in H}(r_i - r_0)d_i \leq 0.$$

For $x \in F_{P\cup\{k\}}$, the carrier calculation in Proposition 5.6 gives $d(S) \geq \gamma_k > 0$. Substitution and division by $D$ prove (RG). □

**Corollary 5.8** (residual-gap escape criterion)**.** *Under the hypotheses of Proposition 5.7, if some $k \in K$ satisfies*

$$\max_{x \in F_{P\cup\{k\}}} \sum_{i \in H} \frac{r_i - r_0}{D}(x_i - \phi_i) \geq 0, \tag{E}$$

*then $k$ satisfies the carrier inequality*

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \leq [\phi_k - v(\{k\})].$$

*Thus existence of a minimum-residual carrier is reduced to escape from one explicit scalar separator; no coordinatewise choice of a member of $K$ is required.*

*Proof.* The contrapositive is exactly inequality (RG). □

The strict separator also has a finite dual form. This removes the quantification over all points of the carrier-tight faces and exposes the precise slack identity induced by the residual cover.

**Proposition 5.9** (dual form of the carrier-face separator)**.** *Use the hypotheses and notation of Proposition 5.6, and put $T_k = P \cup \{k\}$. A vector $w \in \mathbb{R}^S_+$ satisfies (Sep) if and only if, for every $k \in K$, there are multipliers $\theta^k_U \geq 0$ on the nonempty proper coalitions and $a_k, b_k \in \mathbb{R}$ such that*

$$w^S = a_k \mathbf{1}_N + b_k \mathbf{1}_{T_k} - \sum_{\emptyset \neq U \subsetneq N} \theta^k_U \mathbf{1}_U, \tag{D1}$$

*where $w^S$ is $w$ on $S$ and zero on $P$, and*

$$\sum_U \theta^k_U \, [\phi(U) - v(U)] < b_k [\phi(T_k) - v(T_k)]. \tag{D2}$$

*Whenever the carrier marginal fails at $k$, the right-hand slack in (D2) is positive and every such certificate has $b_k > 0$.*

*Proof.* Fix $k$ and maximize $w^S \cdot x$ over the core subject to $x(T_k) = v(T_k)$. In the convention $-x(U) \leq -v(U)$ for the core inequalities, the linear-programming dual has nonnegative multipliers $\theta^k_U$, free multipliers $a_k, b_k$ on efficiency and the face equality, and stationarity exactly (D1). Its objective is

$$-\sum_U \theta^k_U v(U) + a_k v(N) + b_k v(T_k).$$

Taking the scalar product of (D1) with $\phi$ shows that this objective is strictly below $w \cdot \phi_S$ exactly when (D2) holds. Strong duality therefore proves the equivalence, simultaneously for all $k$ because the same $w$ occurs in each stationarity identity.

If the carrier marginal fails, then $\phi(T_k) - v(T_k) > \phi_k - v(\{k\}) \geq 0$. The left side of (D2) is nonnegative, so strictness forces $b_k > 0$. □

**Corollary 5.10** (aggregated separator identity)**.** *If the carrier marginal fails for every $k \in K$, then the separator in Proposition 5.6 supplies $c > 0$, $A \in \mathbb{R}$, a collection $\eta_U \geq 0$, and a nonzero $w \in \mathbb{R}^S_+$ such that*

$$\mathbf{1}_K = c w^S - A \mathbf{1}_N - |K| \mathbf{1}_P + \sum_U \eta_U \mathbf{1}_U, \tag{AD1}$$

*and*

$$\sum_{U} \eta_U[\phi(U) - v(U)] < \sum_{k \in K} [\phi(P \cup \{k\}) - v(P \cup \{k\})]. \tag{AD2}$$

*More precisely, one may take*

$$c = \sum_{k \in K} \frac{1}{b_k}, \qquad A = \sum_{k \in K} \frac{a_k}{b_k}, \qquad \eta_U = \sum_{k \in K} \frac{\theta_U^k}{b_k}.$$

*Proof.* Proposition 5.9 gives $b_k > 0$ for every $k \in K$. Divide (D1) and (D2) by $b_k$ and sum over $k$. Since $\mathbf{1}_{P \cup \{k\}} = \mathbf{1}_P + e_k$ and $\sum_{k \in K} e_k = \mathbf{1}_K$, the stationarity sum rearranges to (AD1); the strict slack inequalities sum to (AD2). □

The same comparison has a quantitative converse at the level of total $S$-payoff. For a core point $\psi$, define

$$\gamma_S(\psi) = \min_{\emptyset \neq C \subsetneq S} \{[\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)]\},$$
$$\eta_S(\psi) = \max\{x(S) - \psi(S) : x \in \mathrm{Core}(v),\ x_i \geq \psi_i\ (i \in S)\}.$$

Thus $(\star_S)$ at $\psi$ is equivalent to $\eta_S(\psi) > 0$.

**Lemma 5.11** (exact carrier lift)**.** *If $v$ is exact, then for every nonempty proper $C \subset S$ there is a $z^C \in \mathrm{Core}(v)$ such that*

$$z^C(S) - \psi(S) \geq [\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)].$$

*Proof.* Choose $z^C$ tight at $P \cup C$. Core feasibility at $C$ gives $z^C(P) \leq v(P \cup C) - v(C)$. Efficiency and the identity $\psi(P) = v(N) - \psi(S)$ now give the displayed inequality. □

The only possible obstruction is coordinatewise: the carrier-tight points raise the total payoff of $S$, but different members of $S$ may fall at different points. For a pair, convexity removes that obstruction without loss.

There is a useful normalization that isolates exactly what remains to be proved and, unexpectedly, removes the leximin selection from the statement.

**Proposition 5.12** (exact-core closure formulation)**.** *Let $v$ be exact, let $\psi \in \mathrm{Core}(v)$, and put $D = \mathrm{Core}(v) - \psi$ and $s_\psi(A) = \psi(A) - v(A)$. Then*

$$s_\psi(A) = -\min_{d \in D} d(A) = \max_{d \in D} d(N \setminus A). \tag{EC}$$

*Consequently*

$$\gamma_S(\psi) = \min_{\emptyset \neq C \subsetneq S} \left\{ \max_{d \in D} d(S \setminus C) + \min_{d \in D} d(C) \right\}, \tag{CG}$$

*whereas*

$$\eta_S(\psi) = \max\{d(S) : d \in D,\ d_i \geq 0\ (i \in S)\}. \tag{CE}$$

*Thus the quantitative carrier inequality is a statement about every translated exact core, not specifically about a leximin point.*

*Proof.* Exactness gives $v(A) = \min_{x \in \mathrm{Core}(v)} x(A)$, hence $s_\psi(A) = -\min_{d \in D} d(A)$. Every $d \in D$ has $d(N) = 0$, so the last quantity equals $\max_{d \in D} d(N \setminus A)$. Substituting $A = P \cup C$ and then $A = C$ gives (CG); translating the defining program for $\eta_S$ gives (CE). □

The closure qualification in this formulation is essential. For the convex hull in $\mathbb{R}^3$ of

$$(0,0,0), (4,2,-12), (-12,14,3), (18,9,-17), (10,17,-9), (4,-2,-5), (15,20,-5),$$

the right side of (CG) is 3, while the largest nonnegative total in the hull is 0. Append to each point a fourth coordinate equal to minus its total and take the lower-envelope exact game. Its full core is the coalitional closure of those generators: the same reference point still has $\gamma_S = 3$, but now $\eta_S = 27$, attained at $(10, 17, 0, -27)$. Thus arbitrary convexity is false by a gap of 3, while exact-core closure repairs this candidate by a gap of 24.

The remaining obstruction has an exact dual that is formally the slack-priced version of a failure cover. Define the common-coordinate margin

$$\kappa_S(\psi) = \max\{z : d \in D,\ d(S) \geq \gamma_S(\psi),\ d_i \geq z\ (i \in S)\}.$$

**Proposition 5.13** (quantitative orthant dual)**.** *The value $\kappa_S(\psi)$ equals*

$$\min \left\{ \sum_T \lambda_T s_\psi(T) - q\gamma_S(\psi) : \begin{array}{l} \lambda_T, q, w_i \geq 0, \quad \sum_{i \in S} w_i = 1, \\ u\mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T = q\mathbf{1}_S + w^S \end{array} \right\}. \tag{OD}$$

*Here $u$ is free and $w^S$ is $w$ on $S$ and zero on $P$. Consequently (Q) is equivalent to $\kappa_S(\psi) \geq 0$, while the stronger uniform lift asks for $\kappa_S(\psi) \geq \gamma_S(\psi)/|S|$.*

*Proof.* Write the translated core inequalities as $-d(T) \leq s_\psi(T)$, the high-total constraint as $-d(S) \leq -\gamma_S(\psi)$, and the coordinate constraints as $-d_i + z \leq 0$. The nonnegative dual multipliers are respectively $\lambda_T, q, w_i$; efficiency has free multiplier $u$. Stationarity in $z$ gives $\sum_i w_i = 1$, stationarity in $d$ gives the displayed incidence identity, and the dual objective is $\sum_T \lambda_T s_\psi(T) - q\gamma_S(\psi)$. Strong LP duality proves (OD). □

**Proposition 5.14** (two-level orthant separators are impossible)**.** *Let $v$ be exact, $\psi \in \mathrm{Core}(v)$, and $\gamma_S(\psi) > 0$. For every nonempty $A \subseteq S$ there is a $d^A \in D$ such that*

$$d^A(S) \geq \gamma_S(\psi), \qquad d^A(A) \geq \gamma_S(\psi). \tag{OI}$$

*Consequently, if $w = a\mathbf{1}_S + b\mathbf{1}_A$ with $a, b \geq 0$, then some $d \in D$ with $d(S) \geq \gamma_S(\psi)$ satisfies $w \cdot d_S \geq (a+b)\gamma_S(\psi)$. In particular, no strictly positive vector with at most two coordinate levels can strictly separate*

$$\{d_S : d \in D,\ d(S) \geq \gamma_S(\psi)\} \quad \textit{from} \quad \mathbb{R}^S_+.$$

*Thus any failure of the quantitative carrier inequality admits a strictly positive separator, but every such separator must have at least three distinct coordinate levels.*

*Proof.* If $A \subsetneq S$, put $C = S \setminus A$ and choose a core point tight at $P \cup C$. For its displacement $d^A$ from $\psi$, (EC) gives

$$d^A(A) = s_\psi(P \cup C) \geq \gamma_S(\psi) + s_\psi(C) \geq \gamma_S(\psi),$$

while Lemma 5.11 gives $d^A(S) \geq \gamma_S(\psi)$. If $A = S$, the latter lemma with any nonempty proper $C \subset S$ supplies the required point. Hence $w \cdot d^A_S = ad^A(S) + bd^A(A) \geq (a+b)\gamma_S(\psi) > 0$ whenever $w \neq 0$.

If the displayed convex set and the nonnegative orthant are disjoint, strong separation supplies a nonzero $w \geq 0$ whose maximum on the first set is strictly negative. Since that set is compact and every one of its points has $S$-sum at least the positive number $\gamma_S(\psi)$, adding a sufficiently small multiple of $\mathbf{1}_S$ makes $w$ strictly positive while preserving strict separation. A strictly positive vector with at most two levels has the form $a\mathbf{1}_S + b\mathbf{1}_A$, contradicting the first part. □

**Proposition 5.15** (codimension-one uniform carrier lift)**.** *Let $v$ be exact, let $\psi \in \mathrm{Core}(v)$, and let $P = N \setminus S = \{p\}$ be a singleton. Put $m = |S|$. Then*

$$x_i = \psi_i + \frac{\gamma_S(\psi)}{m} \quad (i \in S), \qquad x_p = \psi_p - \gamma_S(\psi) \tag{UL}$$

*defines a point of* $\mathrm{Core}(v)$ *whenever $\gamma_S(\psi) > 0$. Consequently*

$$\eta_S(\psi) \geq \gamma_S(\psi),$$

*and every member of $S$ can be raised simultaneously by $\gamma_S(\psi)/|S|$.*

*Proof.* Write $t = \gamma_S(\psi)/m$ and $d = x - \psi$. Efficiency is immediate. If $T \subseteq S$, then $d(T) = t|T| \geq 0 \geq -s_\psi(T)$.

Now let $T = P \cup C$ with $C \subsetneq S$ and put $A = S \setminus C$. If $C$ is nonempty, the definition of $\gamma_S$ gives

$$s_\psi(T) \geq \gamma_S(\psi) + s_\psi(C) \geq \gamma_S(\psi),$$

and hence

$$d(T) = -d(A) = -t|A| \geq -\gamma_S(\psi) \geq -s_\psi(T).$$

It remains only to check $T = P$. Exactness makes $v$ superadditive on disjoint coalitions: a core point tight at $P \cup C$ gives $v(P \cup C) \geq v(P) + v(C)$. Therefore $s_\psi(P \cup C) \leq s_\psi(P) + s_\psi(C)$, and any nonempty proper $C \subset S$ yields

$$\gamma_S(\psi) \leq s_\psi(P \cup C) - s_\psi(C) \leq s_\psi(P).$$

Thus $d(P) = -\gamma_S(\psi) \geq -s_\psi(P)$. Every coalition constraint holds, so $x \in \mathrm{Core}(v)$ and $x(S) - \psi(S) = mt = \gamma_S(\psi)$. □

**Proposition 5.16** (sharp quantitative pair bound and uniform split)**.** *If $v$ is exact, $\psi \in \mathrm{Core}(v)$, and $S$ is a proper two-player coalition, then there is an $x \in \mathrm{Core}(v)$ satisfying*

$$x_i - \psi_i \geq \frac{\gamma_S(\psi)}{2} \quad (i \in S).$$

*In particular, $\eta_S(\psi) \geq \gamma_S(\psi)$, and the coefficient one is sharp.*

*Proof.* The claim is immediate if $\gamma_S(\psi) \leq 0$, so write $\gamma = \gamma_S(\psi) > 0$. Let

$$K = \{d_S : d \in D,\ d(S) \geq \gamma\}.$$

If $K$ did not meet $(\gamma/2)\mathbf{1}_S + \mathbb{R}^S_+$, strong separation would supply a nonzero $w \geq 0$ such that

$$\max_{d_S \in K} w \cdot d_S < \frac{\gamma}{2} \sum_{i \in S} w_i. \tag{PS}$$

Every nonnegative two-dimensional $w$ can be written $a\mathbf{1}_S + b\mathbf{1}_A$ for $a, b \geq 0$ and a nonempty $A \subseteq S$. Proposition 5.14 gives a point of $K$ with weighted value at least $(a+b)\gamma$, whereas the right side of (PS) is $\gamma(2a + b|A|)/2 \leq (a+b)\gamma$, a contradiction. Hence the uniform split exists. The exact lower-envelope equality fixtures show sharpness. □

**Corollary 5.17** (quantitative carrier bound through four players)**.** *For every exact game with $|N| \leq 4$, every core point $\psi$, and every proper nonsingleton $S$, one has*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

*More generally, the inequality holds in every dimension whenever $|S| = 2$ or $|N \setminus S| = 1$. Hence any counterexample requires $|S| \geq 3$, $|N \setminus S| \geq 2$, and therefore at least five players.*

The first ostensibly open dimension can also be closed, although our present proof is a finite exact computation.

**Proposition 5.18** (computer-assisted five-player carrier theorem)**.** *For every exact game with $|N| \leq 5$, every core point $\psi$, and every proper nonsingleton $S$,*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

*In the sole new case $|N| = 5$, $|S| = 3$, the exactness assumption can be weakened to total balancedness (nonemptiness of every subgame core).*

*Proof.* The preceding results leave only $|N| = 5$, $|S| = 3$, and $P = N \setminus S = \{p, q\}$. The case $\gamma_S(\psi) \leq 0$ is immediate. Otherwise translate $\psi$ to zero, scale so that $\gamma_S(\psi) = 1$, and write $s(A) = s_\psi(A)$. In fact, total balancedness suffices: a core point of each subgame $T$, translated by $\psi|_T$ and extended arbitrarily outside $T$ to have zero grand total, gives a vector $d^T$ such that

$$d^T(N) = 0, \qquad d^T(T) = -s(T), \qquad d^T(U) \geq -s(U) \quad (\emptyset \neq U \subseteq T). \tag{X5}$$

Together with $s \geq 0$, $s(\emptyset) = s(N) = 0$, and the six carrier inequalities

$$s(P \cup C) - s(C) \geq 1 \quad (\emptyset \neq C \subsetneq S), \tag{C5}$$

these constraints define a rational polyhedron containing every normalized totally balanced instance under consideration.

The LP defining $\eta_S$ has the slack-linear dual

$$\min_{\lambda, w, \rho \geq 0} \sum_{\emptyset \neq T \subsetneq N} \lambda_T s(T) \tag{E5}$$

subject to

$$\sum_{T \ni p} \lambda_T = \sum_{T \ni q} \lambda_T = 1 + \rho, \qquad \sum_{T \ni i} \lambda_T + w_i = \rho \quad (i \in S).$$

This is a five-equation standard-form polyhedron with 34 nonnegative variables. The verifier enumerates all $\binom{34}{5} = 278{,}256$ candidate bases by exact elimination. They give 263 distinct vertices, or 43 orbits under the $S_3 \times S_2$ player symmetry. For each orbit representative it then minimizes the objective in (E5) over (X5)–(C5). Independently replayed rational dual certificates give only the lower bounds

$$1, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad 2, \quad 3, \quad 4;$$

the minimum is 1 and is attained. Since a slack-nonnegative objective has an optimal vertex in (E5), every normalized instance has $\eta_S \geq 1 = \gamma_S$. The exact enumeration and all 43 rational certificates are in `src/audit_n5_exact_cone_eta_vertices.py` and `results/n5_totally_balanced_cone_eta_vertices.json`. □

The exact-total uniform displacement used in the codimension-one proof does not extend to this case. An exact integer five-player game recorded in the supplementary counterexample artifact has $\gamma_S = 1$, $\eta_S = 2$, and $\kappa_S = 1/2$, but no displacement with $d(S) = 1$ and $d_i = 1/3$ for all $i \in S$. Thus the five-player theorem genuinely needs the freedom to gain more than $\gamma_S$ in total.

The eta dual has a finite combinatorial interpretation that removes the apparently unbounded parameter $\rho$.

Minimal balanced collections and their generation are classical; complete computational catalogs are now available through seven players (Laplace Mermoud et al. 2025). The point here is their target-conditioned appearance as the vertices of the eta dual, followed by optimization over the exact-game cone. The latter cone has an independent facet description in terms of indecomposable minimal semi-balanced systems (Studený and Kratochvíl 2022).

**Lemma 5.19** (eta-dual vertices are minimal balanced collections)**.** *Fix a proper nonsingleton $S \subset N$ and put $P = N \setminus S$. On every nonnegative slack system, the eta dual in (E5) (with the evident degree equation for every member of $P$) has an optimal vertex with*

$$\lambda_S = 0, \qquad \lambda_{\{i\}} = 0 \quad (i \in S). \tag{DV}$$

*After imposing (DV), its vertices are in bijection with the minimal balanced collections on $N$ that contain $S$. Explicitly, adjoining $S$ with coefficient one to a dual point gives*

$$\mathbf{1}_S + \sum_T \lambda_T \mathbf{1}_T + \sum_{i \in S} w_i \mathbf{1}_{\{i\}} = (1+\rho)\mathbf{1}_N. \tag{MB}$$

*Conversely, rescale a minimal balanced collection containing $S$ so that its $S$-coefficient is one, interpret its $S$-singleton coefficients as the $w_i$, and interpret all remaining coefficients except that of $S$ as the $\lambda_T$. In particular, every relevant dual vertex uses at most $|N|$ coalitions after $S$ is adjoined.*

*Proof.* If $a = \lambda_S > 0$, deleting $a$ from $\lambda_S$ and adding $a$ to every $w_i$, $i \in S$, preserves all degree equations and weakly decreases the objective by $as(S)$. Similarly, $\lambda_{\{i\}}$ can be moved into $w_i$ at weakly lower cost. Hence an optimum exists on the face (DV), and a vertex optimum can be chosen there.

On that face the dual degree equations are exactly (MB). Thus the face is affinely isomorphic to the slice, at $S$-coefficient one, of the nonnegative cone of balanced collections containing $S$. A point of this slice is a vertex precisely when it lies on an extreme ray with positive $S$-coefficient: a decomposition into two such rays gives a convex decomposition after normalization, while a nonzero ray with zero $S$-coefficient gives a recession direction and again precludes extremality. The extreme rays of the balanced-collection cone are the minimal balanced collections. Their positive incidence columns are linearly independent, so there are at most $|N|$ of them. Finally, in the reverse construction the normalized coefficient of $S$ is at most one because every member of $S$ has total balanced degree one; hence $\rho \geq 0$ as required. □

The totally balanced strengthening is dimension-sharp. Its first six-player escape nevertheless exposes a reusable exactness identity.

**Lemma 5.20** (three-block exact-closure repair)**.** *Let $N = \{1, \dots, 6\}$, $S = \{1, 2, 3\}$, $P = \{4, 5, 6\}$, and abbreviate $s = s_\psi$. For every exact game and every $\psi \in \mathrm{Core}(v)$,*

$$s(2345) + s(146) + s(56) \geq s(1456) - s(1). \tag{TB}$$

*Consequently the eta-dual supported with unit weight on $2345, 146, 56$ has cost at least $\gamma_S(\psi)$.*

*Proof.* Translate $\psi$ to zero. Choose core displacements $a, b, c$ tight at $5, 45, 1456$, respectively. The incidence identities

$$\begin{aligned} \mathbf{1}_1 + \mathbf{1}_{2345} + \mathbf{1}_{56} &= \mathbf{1}_N + \mathbf{1}_5, \\ \mathbf{1}_5 + \mathbf{1}_{2345} + \mathbf{1}_{146} &= \mathbf{1}_N + \mathbf{1}_{45}, \\ \mathbf{1}_1 + \mathbf{1}_{45} + \mathbf{1}_{146} + \mathbf{1}_{56} &= 2\mathbf{1}_{1456} \end{aligned}$$

and core feasibility give

$$\begin{aligned} s(5) &\leq s(1) + s(2345) + s(56), \\ s(45) &\leq s(5) + s(2345) + s(146), \\ 2s(1456) &\leq s(1) + s(45) + s(146) + s(56). \end{aligned}$$

Substituting the first two bounds into the third and dividing by two proves (TB). The final assertion follows from $s(1456) - s(1) \geq \gamma_S(\psi)$, the carrier inequality for $C = \{1\}$. □

There is an integer totally balanced game for this split with $\gamma_S = 1$, $\eta_S = 0$, and zero-cost eta dual $\{2345, 146, 56\}$. It is not exact; its exact-core closure is the zero game, so its carrier gap also collapses to zero. Thus Lemma 5.20 identifies exactly the crossing constraint supplied by exactness. The game, closure, and incidence proof are independently checked in the supplementary three-block verifier.

The repair is not isolated: it closes the complete first multilevel layer of the six-player eta dual.

**Proposition 5.21** (complete six-player unit-triple layer)**.** *In the split $|S| = |P| = 3$, consider eta-dual solutions with $\rho = 1$, $w = 0$, and three unit-weight supporting coalitions. Equivalently, every target player has support degree one and every outside player has degree two. For every such dual and every exact normalized slack system satisfying the carrier inequalities, its slack cost is at least one. Hence this entire dual layer satisfies the quantitative carrier bound.*

*Proof.* Each target player chooses its unique supporting coalition, while each outside player chooses the unique coalition from which it is absent. Enumerating these two maps and quotienting by $S_3 \times S_3$ gives exactly 16 support orbits. Minimization over the global exact-witness cone gives exact values one for seven orbits and two for nine. Every value is certified by independently replayed rational primal and dual solutions. The complete enumeration is in `src/audit_n6_unit_triple_eta_duals.py`, with the certificates in `results/n6_unit_triple_eta_duals.json`. □

The preceding layer is the sharp boundary of a complete six-player theorem.

**Proposition 5.22** (computer-assisted six-player carrier theorem)**.** *For every exact game with $|N| \le 6$, every core point $\psi$, and every proper nonsingleton $S$,*

$$\eta_S(\psi) \ge \gamma_S(\psi).$$

*Proof.* Corollary 5.17, Proposition 5.18, and the general cases $|S| = 2$ and $|P| = 1$ leave only the six-player splits $(|S|, |P|) = (3, 3)$ and $(4, 2)$. Translate $\psi$ to zero and, when $\gamma_S(\psi) > 0$, scale to $\gamma_S(\psi) = 1$. The global-witness version of (X5), together with slack nonnegativity and the carrier inequalities (C5), is a rational polyhedron containing every such normalized exact game.

By Lemma 5.19, it suffices to inspect the minimal balanced collections containing the canonical $S$. Such a collection has at most six members. For each split, the enumerator therefore checks exactly

$$\sum_{k=2}^{6} \binom{61}{k-1} = 6{,}508{,}883$$

supports: $S$ is fixed and the other members are chosen from the remaining 61 nonempty proper coalitions. Exact determinant tests find 21,695 minimal balanced collections for the $3 + 3$ split and 19,501 for the $4 + 2$ split. Quotienting by $S_3 \times S_3$ and $S_4 \times S_2$ leaves, respectively, 870 and 619 orbits.

For every one of these 1,489 orbit representatives, the corresponding eta objective is minimized over the global exact-witness polyhedron. Independently replayed rational primal and dual certificates prove that all optima are at least one. The exact minimum is one in both splits, attained by seven $3 + 3$ orbits and two $4 + 2$ orbits. This proves $\eta_S \ge 1 = \gamma_S$ in the only remaining cases.

The C++ census is `src/enumerate_n6_eta_dual_vertices.cpp`; its exact output and independent structural verification are `results/n6_eta_dual_vertex_census.json` and `results/n6_eta_dual_vertex_census_verification.json`. The latter rechecks the support count, balancing equations, incidence rank, orbit uniqueness, and recovery of all 505 previously sampled orbits. The exact LP verifier and its full rational certificates are `src/audit_n6_eta_dual_vertices.py` and `results/n6_eta_dual_vertex_audit.json`. Finally, `src/verify_n6_eta_dual_vertex_audit.py` replays all 1,489 saved lower-bound identities in exact arithmetic without calling an optimizer; its summary is `results/n6_eta_dual_vertex_audit_verification.json`. □

The six-player threshold is exact.

**Proposition 5.23** (sharp seven-player exact counterexample)**.** *There is an exact seven-player game whose leximin point $\phi$ and a proper coalition $S$ satisfy*

$$\gamma_S(\phi) = \frac{3}{4}, \qquad \eta_S(\phi) = 0.$$

*For the same game the core reference $\psi = 0$ satisfies $(\gamma_S(\psi), \eta_S(\psi)) = (1, 0)$. Consequently the quantitative carrier inequality is false in unrestricted dimension, and Proposition 5.22 is sharp in the number of players.*

*Proof.* Let $N = \{1, \dots, 7\}$, $S = \{1, 2, 3\}$, $P = \{4, 5, 6, 7\}$, and put

$$a = (-3, 1, 3, -1, 2, -1, -1), \qquad b = (3, -2, 0, -1, -1, -1, 2).$$

Both vectors have total zero. Define the lower-envelope game

$$v(T) = \min\{0, a(T), b(T)\} \qquad (T \subseteq N),$$

and take $\psi = 0$. Each of $0, a, b$ belongs to $\mathrm{Core}(v)$, and at least one of them is tight at every coalition. Hence $v$ is exact and $\psi \in \mathrm{Core}(v)$.

Writing $s(T) = -v(T)$, direct evaluation gives

| $C$ | 1 | 2 | 3 | 12 | 13 | 23 |
|---|---|---|---|---|---|---|
| $s(C)$ | 3 | 2 | 0 | 2 | 0 | 2 |
| $s(P \cup C)$ | 4 | 3 | 1 | 3 | 1 | 3 |

and therefore every one of the six carrier differences equals one. Thus $\gamma_S(0) = 1$.

It remains to calculate $\eta_S(0)$. The four coalitions

$$Q_1 = 13456, \quad Q_2 = 457, \quad Q_3 = 3467, \quad Q_4 = 567$$

all have value zero, and their incidence vectors satisfy

$$\mathbf{1}_S + \sum_{j=1}^{4} \mathbf{1}_{Q_j} + e_1 + 2e_2 = 3\mathbf{1}_N. \tag{N7}$$

If $x \in \mathrm{Core}(v)$ and $x_i \geq 0$ for $i \in S$, then $x(Q_j) \geq v(Q_j) = 0$ for every $j$. Taking the scalar product of (N7) with $x$ yields

$$x(S) + \sum_{j=1}^{4} x(Q_j) + x_1 + 2x_2 = 3x(N) = 0.$$

Every term on the left is nonnegative, so $x(S) = 0$. Since $x = 0$ is feasible, $\eta_S(0) = 0$.

The leximin point can be certified in three elementary stages, without an optimization oracle. The singleton floors are

$$m = (-3, -2, 0, -1, -1, -1, -1).$$

At the first stage, $v(125) = 0$ and efficiency imply $x(3467) \leq 0$ for every $x \in \mathrm{Core}(v)$. If all seven gains $x_i - m_i$ are at least $t$, then

$$x(3467) \geq m_3 + m_4 + m_6 + m_7 + 4t = -3 + 4t,$$

so $t \leq 3/4$. Equality fixes players $3, 4, 6, 7$ at $(3/4, -1/4, -1/4, -1/4)$.

On this first-stage face their coordinates sum to zero. Since $v(13) = v(27) = 0$, the core constraints give $x_1 \geq -3/4$ and $x_2 \geq 1/4$. Efficiency therefore gives $x_5 \leq 1/2$, so the second common gain among the free players is at most $x_5 - m_5 \leq 3/2$; equality fixes player 5 at $1/2$. Finally $x_1 + x_2 = -1/2$. If their remaining two gains are at least $u$, then

$$-\frac{1}{2} = x_1 + x_2 \geq m_1 + m_2 + 2u = -5 + 2u,$$

so $u \leq 9/4$. All three bounds are attained by

$$\phi = \frac{1}{4}a = \left(-\frac{3}{4}, \frac{1}{4}, \frac{3}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}\right).$$

Thus the successive Maschler strata are

$$\{3,4,6,7\} \text{ at } \frac{3}{4}, \qquad \{5\} \text{ at } \frac{3}{2}, \qquad \{1,2\} \text{ at } \frac{9}{4},$$

which proves directly that this point is $\phi(v)$. Direct evaluation of $s_\phi(T) = \phi(T) - v(T)$ gives $s_\phi(P \cup C) - s_\phi(C) = 3/4$ for all six nonempty proper $C \subset S$, so $\gamma_S(\phi) = 3/4$. Moreover each $Q_j$ above is $\phi$-tight. If $x \in \mathrm{Core}(v)$ dominates $\phi$ on $S$, put $d = x - \phi$. Taking the scalar product of (N7) with $d$ gives

$$d(S) + \sum_{j=1}^{4} d(Q_j) + d_1 + 2d_2 = 3d(N) = 0.$$

Every term on the left is nonnegative: the coordinate terms by domination, and the coalition terms because $x(Q_j) \geq v(Q_j) = \phi(Q_j)$. Hence $d(S) = 0$, while $x = \phi$ is feasible, proving $\eta_S(\phi) = 0$.

The construction and all 128 lower-envelope, core, carrier, degree, and zero-cost identities are replayed in integer arithmetic by `src/verify_n7_exact_carrier_counterexample.py`. The exact rational leximin calculation and the six 3/4 differences are independently replayed by `src/verify_n7_exact_carrier_counterexample_leximin.py`. □

The sharp counterexample propagates to every higher dimension.

**Corollary 5.24** (complete dimension threshold)**.** *The quantitative carrier inequality holds for every exact game if $|N| \leq 6$, and fails for an exact game at every dimension $|N| \geq 7$.*

*Proof.* The positive statement is Proposition 5.22. Starting from the game in Proposition 5.23, append any number of null players: extend each of the three generators by zero and put $v^+(T) = v(T \cap N)$. This remains an exact lower-envelope game.

Its core is exactly $\mathrm{Core}(v) \times \{0\}^k$. Indeed, singleton rationality makes each new coordinate nonnegative, while the old-grand-coalition constraint and efficiency make their sum nonpositive; hence all are zero. The converse inclusion is immediate from $v^+(T) = v(T \cap N)$. The new players therefore contribute fixed zero gains before the three positive leximin strata, so the old leximin coordinates are unchanged. For the same target $S$, the enlarged outside set merely adds null players, and hence all six carrier differences remain 3/4. The feasible core displacements on $S$ are also unchanged, so $\eta_S = 0$. The identities are replayed through arbitrary prescribed finite dimension by `src/verify_n7_exact_carrier_dummy_extensions.py`. □

The example has a rigid two-dimensional core but is not an isolated game.

**Proposition 5.25** (triangular core and generator minimality)**.** *For the seven-player game in Proposition 5.23,*

$$\mathrm{Core}(v) = \mathrm{conv}\{0, a, b\}.$$

*Moreover every representation of $v$ as the lower envelope of core allocations requires at least three allocations. Thus the displayed three-generator representation is minimal.*

*Proof.* The eight coalitions

$$\mathcal{R} = \{1234, 125, 1236, 13456, 27, 457, 3467, 567\}$$

all have value zero and satisfy $\sum_{R \in \mathcal{R}} \mathbf{1}_R = 4\mathbf{1}_N$. Hence every core point is tight on every member of $\mathcal{R}$: all eight core inequalities are nonnegative, while their sum is four times efficiency.

Their incidence matrix has rank five. Both $a$ and $b$ lie in its kernel and are linearly independent, so every $x \in \mathrm{Core}(v)$ has the form $x = \alpha a + \beta b$.

The core inequalities at 3, 13, and 4 now read, respectively, $3\alpha \geq 0$, $3\beta \geq 0$, and $-\alpha - \beta \geq -1$. Thus $\alpha, \beta \geq 0$ and $\alpha + \beta \leq 1$. The reverse inclusion holds because $0, a, b \in \mathrm{Core}(v)$ and the core is convex.

Finally, among the three vertices, $a$ uniquely attains $v(1)$, $b$ uniquely attains $v(2)$, and zero uniquely attains $v(123)$. Any lower-envelope representation by core allocations must therefore contain all three. The rank, degree, core, and unique-attainment calculations are replayed by `src/verify_n7_exact_carrier_structure.py`. □

**Proposition 5.26** (nonisolated one-parameter family)**.** *For every $1/4 < \theta \leq 1$, the exact game*

$$v_\theta(T) = \min\{0, a(T), \theta b(T)\}$$

*has leximin point $\phi = a/4$ and, for $S = 123$,*

$$\gamma_S(\phi) = \theta - \frac{1}{4} > 0, \qquad \eta_S(\phi) = 0.$$

*Consequently the seven-player counterexample is neither isolated nor unique up to permutation and common positive scaling.*

*Proof.* Exactness again follows from the equal-total lower-envelope construction, and $a/4$ is a core point. For $1/4 < \theta \leq 1$, the singleton floors are

$$(-3, -2\theta, 0, -1, -\theta, -1, -1).$$

The first leximin stage is unchanged: $v_\theta(125) = 0$ forces $x(3467) \leq 0$, so the first common gain is at most $3/4$ and fixes $3, 4, 6, 7$ at the coordinates of $a/4$. On that face the constraints at 13 and 27 give $x_1 \geq -3/4$ and $x_2 \geq 1/4$, whence efficiency gives $x_5 \leq 1/2$. The next gain is therefore at most $1/2 + \theta$, attained by $a/4$ and no larger than either other free gain because $1/4 < \theta \leq 1$.

After fixing player 5, efficiency gives $x_1 + x_2 = -1/2$. The preceding bound on $x_1$ gives $x_2 \leq 1/4$, so the next gain is at most $1/4 + 2\theta$, again attained by $a/4$; the remaining coordinate is then fixed. This proves the leximin assertion directly. Evaluation of the six carrier differences gives $3/4$ for the three $C$ containing player 1 and $\theta - 1/4$ for the other three. The four coalitions in (N7) remain zero-valued and $\phi$-tight, so the same identity proves $\eta_S(\phi) = 0$.

For example, $\theta = 1$ and $\theta = 4/5$ have singleton-floor multisets that are not proportional under any permutation, proving inequivalence. Exact family checks are in `src/verify_n7_exact_carrier_family.py`. □

**Corollary 5.27** (the carrier failure is not bumpable)**.** *For every game in Proposition 5.26, the exact bump capacity at $S = 123$ is zero:*

$$d^*_{v_\theta}(123) = 0.$$

*Hence this sharp failure of the quantitative carrier inequality cannot itself produce a counterexample to proper-coalition monotonicity.*

*Proof.* The coalition 135 has payoffs 0, 2, and $2\theta$ at the three generators $0, a, \theta b$. It is therefore uniquely tight at zero. Since $v_\theta(123) = 0$ as well,

$$\max_{x \in F_{135}} x(123) - v_\theta(123) = 0.$$

This term occurs in the minimum defining $d^*_{v_\theta}(123)$, while bump capacity is nonnegative, proving equality. Thus no positive exact bump at this target exists. □

The negative mechanism also exposes a rank threshold. After adjoining $S$, the zero-cost eta support in (N7) is a minimal balanced collection of seven linearly independent incidence columns. Such a collection cannot occur on six players. This explains why this particular obstruction first appears at seven, although the positive six-player theorem still uses the complete enumeration to exclude all other minimal-balanced supports of rank at most six.

The slack certificate is automatic on the most important thin-core model class used by the searches.

**Proposition 5.28** (two-generator lower envelopes)**.** *Let $p, q \in \mathbb{R}^N$ have the same total, and define the exact game*

$$v(T) = \min\{p(T), q(T)\} \quad (T \subsetneq N), \qquad v(N) = p(N) = q(N).$$

*Let $S$ be proper. If $\max_{x\in\mathrm{Core}(v)} x(S) > \phi(S)$ and $(\star_S)$ fails at $\phi$, then there is a nonempty proper $C \subset S$ satisfying the hypothesis of Proposition 5.5. Consequently $d^*_v(S) \le \phi(S) - v(S)$.*

*Proof.* Both $p$ and $q$ belong to $\mathrm{Core}(v)$, and one of them attains $v(T)$ at every proper $T$, so $v$ is exact. Put $P = N \setminus S$. Exactness at $P$ gives

$$\max_{x\in\mathrm{Core}(v)} x(S) = v(N) - v(P) = \max\{p(S), q(S)\}.$$

After interchanging $p$ and $q$, binding therefore gives $p(S) > \phi(S)$. The two generator sums on $S$ cannot tie, since then $v(S) = \max_{x\in\mathrm{Core}(v)} x(S)$; hence $p(S) > q(S)$ and $p(P) < q(P)$. Define

$$C = \{i \in S : p_i > \phi_i\}.$$

This set is nonempty because $p(S) > \phi(S)$, and it is proper because otherwise $p \in \mathrm{Core}(v)$ would itself witness $(\star_S)$. For $i \in C$, core feasibility at the singleton gives

$$p_i > \phi_i \ge v(\{i\}) = \min\{p_i, q_i\},$$

so $q_i \le \phi_i < p_i$. Hence

$$p(C) - q(C) \ge p(C) - \phi(C) \ge p(S) - \phi(S) = \phi(P) - p(P).$$

In particular $q(C) < p(C)$, and therefore

$$\begin{aligned} v(P \cup C) - v(C) &= \min\{p(P) + p(C), q(P) + q(C)\} - q(C) \\ &= \min\{p(P) + p(C) - q(C), q(P)\} \\ &\ge \phi(P). \end{aligned}$$

The last inequality also uses $\phi(P) \le v(N) - v(S) = q(P)$, since $q(S) < p(S)$. Equivalently, $\phi(P \cup C) - v(P \cup C) \le \phi(C) - v(C)$, which is precisely the off-side slack-domination hypothesis. □

**Lemma 5.29** (exact-face compensation identity)**.** *Let $v$ be exact, let $P = N \setminus S$, and let $\psi \in \mathrm{Core}(v)$. For nonempty $C \subsetneq S$, choose any $a \in \mathrm{Core}(v)$ with $a(P \cup C) = v(P \cup C)$; such an $a$ exists by exactness. Then*

$$v(P \cup C) - v(C) - \psi(P) = [a(P) - \psi(P)] + [a(C) - v(C)]. \tag{GC}$$

*In particular, $a(P) \geq \psi(P)$ is sufficient for the off-side carrier inequality at $C$; otherwise the only missing quantity is the nonnegative trace compensation $a(C) - v(C)$. Thus the identity is not restricted to a finite lower-envelope representation.*

*Proof.* Since $a(P \cup C) = v(P \cup C)$ and $P \cap C = \varnothing$, the identity is algebraic. Core feasibility gives $a(C) - v(C) \geq 0$. □

So the bump is realized by a continuous path of *exact* games in which only the lower bound $\tau$ on $x(S)$ rises. This isolates the role of exactness: if endpoint exactness is weakened to balancedness, the analogous interior path need not be exact, and that is precisely the room inhabited by the counterexamples of Section 4. For proper nonsingleton bumps on exact games, coalitional monotonicity is therefore equivalent to the one-parameter statement:

**Lemma 5.30** (monotone path — OPEN)**.** *Along the exact path $v_\tau$, the map $\tau \mapsto \phi_i(v_\tau)$ is nondecreasing for every $i \in S$ (and, empirically, nonincreasing for $i \notin S$).*

Two reductions cut the work down further. First, the common case is trivial:

**Lemma 5.31** (non-binding)**.** *If $\phi(v)(S) \geq v(S) + d$ then $\phi(v') = \phi(v)$.*

*Proof.* $\phi(v)$ is the leximin point of $\mathrm{Core}(v) \supseteq \mathrm{Core}(v')$ and lies in $\mathrm{Core}(v')$, hence is the leximin point of the smaller set too; $\phi$ is single-valued. □

Binding ($\phi(v)(S) < v(S) + d$) requires $\phi(v)(S) < \max_{\mathrm{Core}} x(S)$ and is empirically rare. In every binding case observed (thousands of exact-preserving bumps at $n \leq 6$, zero violations) one finds $\phi(v')(S) = v(S) + d$ and the two-sided pattern $\phi_i(v') \geq \phi_i(v)$ for $i \in S$, $\leq$ for $i \notin S$.

**A cone criterion.** For $x = \phi(w)$ with tight set $\mathcal{T}(x) = \{T : x(T) = w(T)\}$, say $(\star_S)$ holds if there is $e \in \mathbb{R}^N$ with

$$e(N) = 0, \quad e(S) > 0, \quad e_i \geq 0 \ \ (i \in S), \quad e(T) \geq 0 \ \ (T \in \mathcal{T}(x)) :$$

a first-order direction raising $x(S)$ without lowering any member of $S$ or breaking a tight core constraint. Empirically $(\star_S)$ is *necessary* for the bump to be monotone (it fails at $\phi(v)$ in Example 4.2, which is why player 2 must drop), and exactness supplies it. This suggests the skeleton

(A) Lemma 5.1: the bump path is exact;

(D) each exact $v_\tau$ satisfies $(\star_S)$ at $\phi(v_\tau)$;

(E′) $(\star_S)$ along the whole path implies Lemma 5.30,

whose composition is the theorem. The supporting evidence is summarized in Table 1; (D) and (E′) remain open, and $(\star_S)$ alone is *not* sufficient (the last row), so a proof of (E′) must use exactness along the whole path, not merely $(\star_S)$ at one endpoint.

Table 1: Computational evidence (float LP, exact-certifiable) for the reduction. $(\star_S)$ evaluated at $\phi$ of the base game; "binding" bumps only.

| statement | sample | violations |
|---|---|---|
| $\phi$ monotone for sampled proper exact bumps (mixed sizes) | 2000+ bumps, $n \le 6$ | 0 |
| (D) exact $\Rightarrow (\star_S)$ at $\phi$ | 39/39, 13/13 | 0 |
| monotone $\Rightarrow (\star_S)$ | all balanced | 0 |
| $(\star_S)$ along the path $\Rightarrow$ monotone | 600 | 0 |
| $(\star_S)$ at one endpoint $\Rightarrow$ monotone | 1696 balanced | 22 non-exact |

**The open problem, sharply.** A Farkas dual turns (D) into a statement about a single maximal face of the core. For a core point $y$ written as a mixture of core points, $y(T) \ge v(T) = \phi(T)$ at every tight $T$, so $e = y - \phi$ meets the tight-cone part of $(\star_S)$ automatically; hence

$$\begin{aligned}(\star_S) &\iff \exists\, y \in \mathrm{Core}(v) :\ y_i \ge \phi_i \ (i \in S) \text{ and } y(S) > \phi(S)\\ &\iff \max\{y(S) : y \in \mathrm{Core}(v),\ y_i \ge \phi_i \ (i \in S)\} > \phi(S).\end{aligned}$$

Dualizing (verified in 200/200 random balanced cases, exact-certifiable):

> $(\star_S)$ *fails* at $\phi(v)$ if and only if $\phi(v)$ maximizes
>
> $$\sum_{k \in S} r_k\, x_k$$
>
> over $\mathrm{Core}(v)$ for some weights $r_k \ge 1$ on $S$ and zero weights outside $S$.

Equivalently, the tight coalitions $\{T : \phi(T) = v(T)\}$ carry nonnegative weights and a $\beta \ge 1$ covering each $k \notin S$ exactly $\beta$ and each $k \in S$ at most $\beta - 1$. Thus **Lemma 5.30's hypothesis (D) is exactly the assertion that an exact game's leximin point never lies on a positive-$S$-weighted maximal face of its core at a binding-bumpable $S$.** Uniform weights $r_k \equiv c$ are already excluded, since maximizing $c\,x(S)$ would force $\phi(S) = \max_{\mathrm{Core}} x(S)$, contrary to binding-bumpability; *the entire open content is that exactness excludes the non-uniform such faces.* The non-exact counterexamples of Section 4 realize exactly such faces—this is why $(\star_S)$ fails there and a member of $S$ drops. (A corner scan for a deficient $S$-member pinned at its core-maximum found no occurrence in any exact binding case, $0/16^{+}$.)

**The natural certificate is provably wrong.** Where $(\star_S)$ *does* fail, one would like to prove $d_v^*(S) = 0$ and thereby certify directly that $S$ is not exactly bumpable. This target is false. From the failure certificate, take the cover $\{\lambda_U\}$ dual to the failure, $W = \bigcup\{U : \lambda_U > 0\}$, and propose $C = W \cap S$. This route is closed by a six-player exact game (supplement, `WITNESS_KEY_FALSE_n6`): $(\star_S)$ fails at $S = \{1,3,4,5\}$ with cover $\{2,3\},\{1,2,6\},\{3,4,6\}$, yet a core point $x$ with $x(W \cap S) = v(W \cap S)$ attains $x(S) = 24 > \phi(S) = 71/3$, so $W \cap S$ certifies nothing; the associated slack inequality $\phi(W) - v(W) \le \phi(W \cap S) - v(W \cap S)$ fails by $1/3$. Such failures are rare ($\approx 0.1\%$ of $(\star_S)$-failures at $n \in \{6,7\}$, found only under generators biased toward thin cores with nested tight chains), which is why sampling first suggested the cover certificate was universal. In every observed failure, however, the operative bound survives: there is a coalition $C$ with $M^{(C)} := \max\{x(S) : x \in \mathrm{Core}(v),\ x(C) = v(C)\} \le \phi(S)$, so that $S$

cannot be bumped past $\phi(S)$ while preserving exactness (in a $\approx 0.2\%$ minority $d_v^*(S) > 0$ — exact six-player witnesses with $d^* = 1$ and $d^* = 2$ are included in the supplement — but always $d_v^*(S) \le \phi(S) - v(S)$). Empirically the certificate has sharp structure: $C$ can always be taken as a subset of $S$ contained in a *single stratum* of the leximin computation — almost always a singleton. If $C$ is $\phi$-tight, then $\phi \in F_C$, so a certificate can only be boundary-sharp, $M^{(C)} = \phi(S)$; exact boundary cases of this kind do occur, while every strict certificate is necessarily non-tight. The open problem is thus sharpened: produce, from the structure of a $(\star_S)$-failure at the leximin point of an exact game, a single-stratum $C \subseteq S$ with $M^{(C)} \le \phi(S)$. Empirically the certificate is pinned further: writing $r$ for the residuals of the failure's dual cover ($r_i \ge 1$ on $S$), a *minimum-residual* member $k \in S$ (one with $r_k = 1$) certifies in 99.6% ($n = 6$) to 100% ($n = 7$) of observed failures, the residue requiring a same-stratum pair. By strong duality the singleton form is equivalent to exhibiting nonnegative weights on $\phi$-tight coalitions with coverage $\beta$ on $N \setminus S$, $\beta - 1$ on $S \setminus \{k\}$, and at least $\beta - 1$ on $k$; weights supported inside $S$ suffice in only about two thirds of cases, so a proof must construct this dual cover from $\phi$-tight coalitions *crossing* $S$. The residual normal gives a possible stronger route, but not an equivalent reformulation. For a minimum-residual $k$, normalized by $r_k = 1$, and any core $x$ with $x_k = v(\{k\})$, maximality of $\phi$ gives

$$[x(S) - \phi(S)] + \sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \le 0.$$

Consequently the singleton face certificate would follow from the sufficient inequality

$$\sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \ge 0$$

throughout that face. The sign is essential; the opposite inequality gives no upper bound on $x(S) - \phi(S)$. Whether this stronger face inequality is forced by the leximin strata is a separate question from the carrier route. An exact audit answers that stronger question negatively. A highest-gain minimum-residual player satisfies the nonnegative-correction condition in 139/141 saved multilevel failures, but not in two seven-player cases. In one of them (`WITNESS_KEY_CX_n7_s303_1491`) the correction is identically $-3/2$ on each of the two highest-gain singleton faces, although both faces certify and both carrier inequalities hold. Thus the residual normal alone cannot prove the multilevel branch; a proof must use additional exactness geometry such as the off-side carrier comparison.

A new proof-producing exact audit isolates a still sharper target on the saved hard frontier. It covers all 141 stored three-or-more-level failures (106 at $n = 6$ and 35 at $n = 7$). In every case some minimum-residual singleton certifies, and, more strongly, some minimum-residual $k$ satisfies the elementary carrier inequality

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \le [\phi(\{k\}) - v(\{k\})].$$

Across the 367 individual minimum-residual candidates, 282 singleton faces certify and the carrier inequality explains 280 of them; it never produces a false certificate. The remaining two seven-player singleton certificates require a richer face dual, but another minimum-residual player in the same case has the carrier certificate. This is exact finite evidence, not a theorem. Nor can it be promoted to an unconditional singleton-carrier conjecture: the exact two-level witness above has no certifying singleton; its minimum-residual pair is essential. At this stage the apparent frontier was a dichotomy. A failure dual with at most two residual levels is already handled by

Proposition 5.3. On the genuinely three-or-more-level branch, these data first suggested proving the displayed singleton inequality. The whole-class audit below supersedes that target with an existential full minimum class $K$ and removes the singleton-selection problem. Proposition 5.46 below will show that the failure cover, and hence $K$, still has to be selected. The saved corpus initially suggested a canonical gain selector: in all 141 cases a carrier-certifying $k$ could be chosen to maximize $\phi_i - v(\{i\})$ inside the minimum-residual class. That refinement is false. A seeded mutation audit produced 534 further exactly certified multilevel records (505 distinct game–coalition pairs). Some minimum-residual carrier still exists in all 534 records, but a highest-gain carrier exists in only 520, and a highest-gain singleton face certifies in only 533.

Here is the sharp selector counterexample. Let the exact six-player game be the lower envelope of the four common-sum points

$$p = (-6, 4, 18, -4, 11, -10), \quad q = (-1, 4, 24, -4, 0, -10),$$
$$a = (-1, -2, 19, -4, 6, -5), \quad b = (-6, 0, 20, 0, 9, -10),$$

and take $S = \{1, 2, 3, 4, 6\}$. Its leximin point is $\phi = (-3, 0, 20, -2, 6, -8)$, the normalized failure cover has three residual levels and minimum-residual class $K = \{1, 2, 3\}$, and the singleton gains on $K$ are $(3, 2, 2)$. Thus player 1 is the unique highest-gain candidate. Nevertheless its carrier slack is $4 > 3$, and more strongly

$$\max\{x(S) : x \in \mathrm{Core}(v),\ x_1 = v(\{1\})\} = 8 > 7 = \phi(S),$$

so even its singleton face fails. Players 2 and 3 both satisfy the carrier inequality with equality and certify. Tight-hypergraph centrality and cover-weighted adjacency refinements also fail on exact mutations. Any singleton route must therefore remain existential: select *some* minimum-residual player, without imposing a gain ranking. After adding fresh three-, four-, and five-generator screens and deduplicating all corpora, this existential carrier statement holds for all 669 distinct exact multilevel game–coalition pairs (702 certified records). The whole-class carrier statement below supersedes the need to solve this singleton-selection problem.

Proposition 5.6 sharpens what remains to be proved. With $K = \{k \in S : r_k = 1\}$, a counterexample would produce simultaneously the positive-$S$ failure normal $r$ and carrier failure at every member of $K$. Proposition 5.7 shows that the resulting common separator is not a second obstruction to eliminate: it is automatically the normalized residual gap on $S \setminus K$. Thus mere coexistence of the two normals cannot yield a contradiction. A tempting coordinatewise strengthening is also false: exactness and the leximin strata need not permit a convex combination of the carrier-tight faces whose average displacement is nonnegative at every higher-residual coordinate. In the exact six-player instance `WITNESS_KEY_CX_n6_s201_10919`, the best common coordinatewise margin over all such mixtures is $-1/20$, although players 4 and 5 both satisfy the carrier inequality (each with margin $-3/4$).

The scalar condition in Corollary 5.8 is the minimal sufficient statement for a *fixed* failure cover, and it initially survives all 141 saved hard multilevel failures. Across their 367 minimum-residual faces, 277 escape; all 87 failed carriers are strictly negative, while three valid carriers do not escape but another candidate does in the same record.

Adversarial mutation shows that even this fixed-cover target is not universal. Among 505 distinct exact mutation cases, all have a minimum-residual carrier but only 123 have an escape face for the cover returned by the exact failure dual. Already trial 0 of the first seeded face-selector mutation audit has $K = \{1, 2, 3\}$ and normalized residual-gap weights $(2/3, 1/3)$ on players $5, 6$. Players 1 and 2 satisfy the carrier inequality, with margins $-2$ and 0, yet their face maxima

are both $-1/3$; player 3 fails the carrier inequality and has maximum $-4/3$. Consequently neither coordinatewise neutralization nor escape for an arbitrarily chosen failure cover can be the missing theorem. A residual-based proof would have to select a favorable cover from the full dual face jointly with the carrier candidate; otherwise it must return to a cover-independent exactness exchange.

There is one dimension in which this cover-selection problem disappears.

**Lemma 5.32** (three-player residual-order rigidity)**.** *Suppose $|S| = 3$ and $(\star_S)$ fails at a core point $\psi$. If no failure cover has at most two residual levels on $S$, then every failure cover has three distinct residuals, and all failure covers induce the same strict order on $S$. In particular, their minimum-residual class is the same singleton.*

*Proof.* Before the harmless positive normalization, failure covers form a polyhedral cone; the normalized residual set is convex. A residual tie on a three-player set has at most two distinct levels, so every residual vector in the stated branch has a strict order. If two covers induced different orders, their line segment would remain in the failure-cover set. Along that segment the order could change only at a residual tie, producing a cover with at most two levels, a contradiction. □

**Lemma 5.33** (three-player block-star and residual-gap alternatives)**.** *Let $\mathcal{T}(\psi)$ be the proper coalitions tight at a core point $\psi$ and put*

$$\mathcal{D}_\psi = \{d \in \mathbb{R}^N : d(N) = 0,\ d(T) \geq 0 \text{ for every } T \in \mathcal{T}(\psi)\}.$$

*For a pair $\{a, b\} \subset S$ with third player $c$, no failure cover has $r_a = r_b$ if and only if some $d \in \mathcal{D}_\psi$ satisfies*

$$d_c \geq 0, \qquad d_a + d_b \geq 0, \qquad d(S) > 0. \tag{BS}$$

*Consequently, when $|S| = 3$, every failure cover has three distinct residuals if and only if all three singleton–complementary-pair versions of (BS) hold.*

*Now suppose failure covers exist and every cover has the fixed order $r_k < r_i < r_j$. Normalize by $r_k = 1$ and define*

$$\delta = \min\{r_j - r_i : r \text{ is a failure-cover residual vector},\ r_k = 1\}. \tag{AG}$$

*Then $\delta > 0$, and there is $d \in \mathcal{D}_\psi$ with*

$$(d_k, d_i, d_j) = (\delta, 1, -1). \tag{GD}$$

*In particular, if $P = N \setminus S$ and $A = P \cup \{k\}$, then $d(P) = -\delta$ and $d(A) = 0$.*

*Proof.* For every failure cover and every $d \in \mathcal{D}_\psi$, the cover identity gives

$$\sum_{p \in S} r_p d_p = \mu d(N) - \sum_{T \in \mathcal{T}(\psi)} \lambda_T d(T) \leq 0. \tag{P}$$

If $r_a = r_b$, the left side is $r_a(d_a + d_b) + r_c d_c$, which is strictly positive under (BS). This proves one direction. Conversely, apply Farkas' lemma to the normalized failure-cover system with the additional equality $r_a = r_b$. The player-equation multipliers, with sign reversed, give $d \in \mathcal{D}_\psi$; the multipliers of the lower bounds on $r_c$ and on the tied pair give respectively $d_c \geq 0$ and $d_a + d_b \geq 0$. Normalizing the Farkas right-hand side gives $d(S) > 0$. This proves the alternative.

For the second claim, the polyhedron in (AG) is nonempty. Its objective is nonnegative and attains its minimum. If the minimum were zero, an optimizer would have $r_i = r_j$, contrary to the fixed strict order; hence $\delta > 0$. Dualize this LP. The dual player multipliers again define a vector in $\mathcal{D}_\psi$. At every feasible point $r_j > r_i > r_k = 1$, so the lower bounds on $r_i$ and $r_j$ are slack; complementary slackness and the objective $r_j - r_i$ give $d_i = 1$ and $d_j = -1$. The dual price of the normalization $r_k = 1$, together with equality of the primal and dual values, gives $d_k = \delta$. Finally $d(S) = \delta$ and $d(N) = 0$, so $d(P) = -\delta$ and $d(A) = 0$. □

Thus the unresolved $|S| = 3$ branch has no chamber-choice ambiguity: after Proposition 5.3, it reduces to proving the carrier inequality for one uniquely determined minimum-residual player. This is strictly simpler than the four-player geometry below, where the residual cone can cross several genuinely multilevel chambers.

Lemma 5.33 replaces the universal quantifier over the cover polyhedron by three primal tangent directions. It also supplies one canonical direction parallel to the trace face $A = P \cup \{k\}$. In the six exact hard cases, independently verified rational Farkas certificates exist for all 18 pair-tie systems. Moreover every normalized adjacent gap is $\delta = 1$, and the six directions in (GD) are integral. These values are finite evidence; the block-star and residual-gap alternatives themselves are proved for every three-player target.

There is a natural stronger three-player target. If $k$ is that unique minimum-residual player, ask whether

$$\max\{x(P) : x \in \mathrm{Core}(v),\ x(P \cup \{k\}) = v(P \cup \{k\})\} \geq \phi(P). \tag{TF}$$

Any maximizer in (TF) is an exact-face witness in Lemma 5.29, so both terms on the right side of (GC) are nonnegative and the singleton carrier inequality follows. The initial exact chamber audit contained six genuinely multilevel $|S| = 3$ cases. Each had exactly one accessible minimum-class chamber, a strict carrier margin between $-8$ and $-1/2$, and an exact rational primal–dual certificate for (TF). A targeted seven-player screen of 1,976 nondegenerate lower-envelope games found 649 binding star failures and seven float multilevel candidates. Four extremals were recomputed exactly: one collapsed to the proved two-level branch, while the other three supplied three of the six cases above. These counts are evidence; Lemma 5.32 is the proved dimension reduction.

Those six face duals share a sharper pattern. Each has zero grand-coalition price, and every positively weighted core constraint $U$ satisfies $U \cap S = \{k\}$. If $b > 0$ is minus the price of the face equality and $\lambda_U \geq 0$ are the resulting core weights, stationarity reads

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}},$$

and (TF) is exactly the weighted slack inequality

$$\sum_U \lambda_U[\phi(U) - v(U)] \geq b[\phi(P \cup \{k\}) - v(P \cup \{k\})]. \tag{TS}$$

This common dual support pattern was exact finite evidence, not a theorem. The counterexample below has a positive dual coalition disjoint from $S$ and therefore also falsifies the exact-trace-$\{k\}$ support refinement. The next reduction remains valid, but its proposed sign conclusion will likewise be falsified.

Exactness supplies a direct face-coordinate reduction. It neither assumes the observed trace-face dual nor selects a dual representative.

**Lemma 5.34** (three-player residual-spread bound)**.** *Let $v$ be exact and suppose a normalized failure cover has residuals $r_k = 1 < r_i = \rho < r_j = \sigma$ on $S = \{k, i, j\}$, and put $A = P \cup \{k\}$. For every $a \in F_A$,*

$$a(P) - \phi(P) \geq (\rho - 1)[\phi(A) - v(A)] + (\sigma - \rho)(a_j - \phi_j). \tag{FC}$$

*Consequently, writing $M_j^A = \max_{a \in F_A} a_j$, the target margin satisfies*

$$\begin{aligned} \max_{x \in F_A} x(P) - \phi(P) &\geq (\rho - 1)[\phi(A) - v(A)] \\ &\quad + (\sigma - \rho)(M_j^A - \phi_j) \\ &\geq (\rho - 1)[\phi(A) - v(A)] \\ &\quad - (\sigma - \rho)[\phi_j - v(\{j\})]. \end{aligned} \tag{RS}$$

*In particular,* (TF) *holds whenever the right side of* (RS) *is nonnegative.*

*Proof.* Write the tight-coalition failure cover as

$$(1, \rho, \sigma)_S = \mu \mathbf{1}_N - \sum_T \alpha_T \mathbf{1}_T, \qquad \alpha_T \geq 0,$$

where $\phi(T) = v(T)$ whenever $\alpha_T > 0$. For $a \in F_A$, put $\Delta = a - \phi$ and $s(A) = \phi(A) - v(A)$. Efficiency and tightness at $A$ give

$$\Delta(P) + \Delta_k = -s(A), \qquad \Delta_i + \Delta_j = s(A).$$

Pairing the failure cover with $\Delta$, and using core feasibility of $a$ on every positively weighted $\phi$-tight coalition, gives

$$\Delta_k + \rho \Delta_i + \sigma \Delta_j = -\sum_T \alpha_T [a(T) - v(T)] \leq 0.$$

Substituting $\Delta_i = s(A) - \Delta_j$ and $\Delta(P) = -s(A) - \Delta_k$ proves (FC). Maximize $a_j$ over $F_A$. Finally every core point has $a_j \geq v(\{j\})$, which gives (RS). □

The singleton-floor comparison in (RS) is strict in only two of the original six exact three-level cases and has lower bounds $-5/4, -5/3, -3, -2$ in the other four. The stronger face-coordinate bound (FC) proves all six. Its exact values are

$$\frac{1}{2}, \quad 1, \quad \frac{11}{4}, \quad \frac{4}{3}, \quad 0, \quad 2,$$

versus actual target margins $1/2, 1, 15/4, 4/3, 5, 2$; hence it is sharp in four cases. A natural attempted generalization was the one-coordinate inequality

$$M_j^A - \phi_j \geq -\frac{\rho - 1}{\sigma - \rho}[\phi(A) - v(A)]. \tag{HC}$$

Unlike trace extension itself, (HC) asks only for the high-residual coordinate on one exact face. It is nevertheless false.

**Proposition 5.35** (genuinely multilevel face-target counterexample)**.** *Let $v$ be the lower envelope of the three common-sum allocations*

$$\begin{pmatrix} 14 & -11 & 12 & -16 & 9 & 14 & 10 \\ 15 & -11 & 12 & -16 & 8 & 14 & 10 \\ 18 & -6 & 6 & -17 & 6 & 13 & 12 \end{pmatrix}.$$

*Then $v$ is an exact seven-player game. For $S = \{2, 3, 5\}$ its leximin point is*

$$\phi = \left(\frac{47}{3}, -9, \frac{29}{3}, -\frac{49}{3}, \frac{23}{3}, \frac{41}{3}, \frac{32}{3}\right).$$

*The coalition is binding by $5/3$, $(\star_S)$ fails, and every failure cover has three distinct residuals in the order*

$$r_5 < r_3 < r_2.$$

*Nevertheless, for $P = N \setminus S$ and $A = P \cup \{5\}$,*

$$\max_{x \in F_A} x(P) = 23 < \frac{71}{3} = \phi(P). \tag{FCX}$$

*For the normalized cover $(r_5, r_3, r_2) = (1, 2, 3)$ one has $s(A) = 1/3$ and $M_2^A - \phi_2 = -1$, so both (TF) and (HC) fail by $2/3$. Trace extension itself still holds with gap zero, and the desired carrier inequality still holds with margin $s(A) - s(\{5\}) = -4/3$.*

*Proof.* All three rows have sum 32 and are core points of their lower envelope. Every proper-coalition worth is attained by a row, proving exactness. Exact leximin LPs give successive surplus levels $2/3, 5/3, 2, 11/3$, pinning respectively $\{4, 6, 7\}, \{1, 5\}, \{2\}, \{3\}$. A core LP gives the stated binding margin. The tight-cover weights $1, 1, 1, 2$ on coalition masks $25, 49, 69, 104$, with grand coefficient 3, give residuals $(r_2, r_3, r_5) = (3, 2, 1)$ and certify star failure. Exact Farkas certificates make each of the three pair-tie cover systems infeasible, so the whole cover face is genuinely multilevel by Lemma 5.32.

On $F_A$, the point $(15, -10, 11, -16, 8, 14, 10)$ attains both $x(P) = 23$ and $x_2 = -10$. For the $P$-maximum, unit core-dual weights on masks $25, 49, 80, 104$, grand price zero, and face price $-3$ certify 23. A coordinate dual certifies $M_2^A = -10$: its core weights on masks $25, 49, 69, 80, 104$ are respectively $2, 2, 1, 1, 3$, with grand price $-1$ and face price $-4$. Direct substitution gives the two margins in the statement. The restricted $A$-core has the same $P$-maximum 23, proving zero trace gap, while the singleton and carrier slacks give $-4/3$. □

The failure is not a numerical knife edge. Replace the second generator by

$$p^2(t) = (17 - t, -11, 12, -16, 6 + t, 14, 10).$$

Exact recomputation at

$$t = 0, \frac{1}{2}, 1, \frac{4}{3}, \frac{3}{2}, 2, 3$$

keeps the leximin point, binding margin $5/3$, star opening zero, strict three-level residual order, carrier margin $-4/3$, and trace gap zero fixed. The face-target margins are respectively

$$\frac{4}{3}, \quad \frac{5}{6}, \quad \frac{1}{3}, \quad 0, \quad -\frac{1}{6}, \quad -\frac{2}{3}, \quad -\frac{2}{3}.$$

Thus an exact sum-preserving motion within $A$ crosses the stronger target without moving the carrier comparison at all. These rational samples are a path audit, not a claim about every real parameter value.

The failed one-coordinate target has an exact dual form that separates the face price from coalition-slack compensation and diagnoses the crossing.

**Proposition 5.36** (high-coordinate slack–price identity)**.** *Let $F_A$ be nonempty, put $s(C) = \phi(C) - v(C)$, and let $M_j^A = \max_{a \in F_A} a_j$. Every optimal dual for this coordinate maximum can be written with $\lambda_C \geq 0$ and $u, b \in \mathbb{R}$ as*

$$-\mathbf{1}_{\{j\}} = \sum_{\emptyset \neq C \subsetneq N} \lambda_C \mathbf{1}_C + u\mathbf{1}_N + b\mathbf{1}_A, \tag{CD}$$

*and satisfies the exact identity*

$$M_j^A - \phi_j = \sum_C \lambda_C s(C) + b\, s(A). \tag{SP}$$

*Consequently, for any $c \geq 0$,*

$$M_j^A - \phi_j \geq -cs(A) \quad \Longleftrightarrow \quad \sum_C \lambda_C s(C) + (b+c)s(A) \geq 0. \tag{DP}$$

*The equivalence holds for one, hence for every, optimal coordinate dual. In particular $b \geq -c$ is sufficient; if $b < -c$, only the positively weighted coalition slacks can supply the missing amount.*

*Proof.* Write the core inequalities as $-a(C) \leq -v(C)$ and minimize $-a_j$ subject also to grand efficiency and $a(A) = v(A)$. Reversing the signs of the nonpositive inequality-dual variables gives (CD), while strong duality gives

$$-M_j^A = \sum_C \lambda_C v(C) + u\, v(N) + b\, v(A).$$

Pair (CD) with $\phi$, use $\phi(N) = v(N)$, and subtract the dual objective identity. This proves (SP); adding $cs(A)$ proves (DP). Its left side is independent of the chosen optimal dual, which proves the final assertion. □

In the original six exact hard faces, maximizing the face price over the optimal dual face gives

$$-1, \quad -\frac{2}{3}, \quad 0, \quad -2, \quad 0, \quad 0.$$

Here $c = (\rho - 1)/(\sigma - \rho) = 1$. Thus the price condition alone proves five cases. In the fourth, the maximal price is $-2$, but the weighted slack is $5/3$ and $s(A) = 1/3$, so the right side of (DP) is $4/3 > 0$. In Proposition 5.35, the maximal price is $-4$, the weighted slack is $1/3$, and $s(A) = 1/3$; hence the same right side is $-2/3$. Thus the identity distinguishes genuine compensation from a genuine sign failure, while a face-price bound by itself was already false on the positive original corpus.

The closed-form spread bound is one feasible point of a stronger finite exactness optimization. Write $s(C) = \phi(C) - v(C)$ and, for a trace-face dual $(\lambda, b)$, define

$$\begin{aligned} \mathcal{B}(\lambda, A) = \sup\{(c-b)s(A) - \sum_C \eta_C s(C) : \eta_C \geq 0,\ c \geq b,\ q \in \mathbb{R}, \\ \sum_U \lambda_U \mathbf{1}_U + \sum_C \eta_C \mathbf{1}_C - c\mathbf{1}_A = q\mathbf{1}_N\}. \end{aligned} \tag{OB}$$

**Proposition 5.37** (one-face exact-balance bound)**.** *Under the hypotheses of Lemma 5.34, suppose in addition that the face LP has a zero-grand dual $(\lambda, b)$ satisfying*

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}}.$$

*Then*

$$\max_{x \in F_A} x(P) - \phi(P) \geq \mathcal{B}(\lambda, A).$$

*The singleton-floor residual-spread right side in (RS) is obtained by a feasible choice in (OB).*

*Proof.* Fix a feasible $(\eta, c, q)$ and choose, by exactness, $a \in \mathrm{Core}(v)$ with $a(A) = v(A)$. Dot the incidence identity in (OB) first with $a$ and then with $\phi$. Core feasibility for every positively weighted coalition and tightness at $A$ give

$$\sum_U \lambda_U s(U) + \sum_C \eta_C s(C) - cs(A) \geq 0.$$

Subtracting $bs(A)$ instead of $cs(A)$ proves the asserted bound and taking the supremum completes the first claim. For the second, use the tight failure-cover weights as part of $\eta$, add $(\sigma - \rho)$ on the singleton $\{j\}$, and take $c = b + \rho - 1$ and $q = \mu - \rho$. Feasibility is the incidence identity

$$\sum_U \lambda_U \mathbf{1}_U + \sum_T \alpha_T \mathbf{1}_T + (\sigma - \rho)\mathbf{1}_{\{j\}} - (b + \rho - 1)\mathbf{1}_A = (\mu - \rho)\mathbf{1}_N.$$

□

The exact LP for (OB) gives, in the six known three-level cases, respectively

$$\frac{1}{2}, \quad 1, \quad \frac{15}{4}, \quad -\frac{5}{3}, \quad -2, \quad 2.$$

It therefore proves the full face target in four cases and equals the actual target margin in all four, including the case where the closed-form spread bound is $-5/4$ but the target margin is $15/4$. Restricting the auxiliary weights $\eta$ to $\phi$-tight coalitions is infeasible in all six cases, so non-tight slack terms are indispensable. The last two actual target margins exceed their optimal one-face bounds by 3 and 7, but neither remains an unexplained finite case. In the latter, $s(A) = 0$, so $\phi \in F_A$ proves the face target immediately. In the former, a common-sum generator attaining $v(A)$ has outside sum $25 > 71/3 = \phi(P)$ and is the required exact-face witness. Thus all six cases now have proof-producing decompositions; only the last generator argument was specific to the finite lower-envelope model. The unconditional face-coordinate bound above now supplies exact-face proofs for both remaining cases as well, so none of the six finite proofs requires a model-specific generator argument.

The remaining dual-existence question has a clean primal form. For $A = P \cup \{k\}$ write

$$\mathrm{Core}_A(v) = \{y \in \mathbb{R}^A : y(A) = v(A),\ y(B) \geq v(B) \text{ for every } B \subsetneq A\}.$$

The projection of $F_A$ is contained in $\mathrm{Core}_A(v)$, so always

$$\max_{x \in F_A} x(P) \leq \max_{y \in \mathrm{Core}_A(v)} y(P). \tag{TE}$$

Equality is precisely the existence of an optimal restricted allocation that extends to the full core. The corresponding dual statement is exact.

**Lemma 5.38** (trace-extension dual equivalence)**.** *Equality holds in* (TE) *if and only if the full face LP has an optimal dual with zero grand-coalition price. Every such dual can be chosen with all positively weighted core constraints contained in $A$ and has*

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}}$$

*for some $b \geq 1$.*

*Proof.* The restricted problem retains only the coordinates and core constraints inside $A$, together with efficiency on $A$. Any restricted dual lifts to the full face dual by assigning zero price to grand efficiency and zero weight to every other core constraint. Conversely, stationarity for a zero-grand full dual has zero coverage on $S \setminus \{k\}$. Since core weights are nonnegative, every positively weighted coalition is contained in $A$; deleting the unused coordinates gives a restricted dual of the same value. Strong duality proves the equivalence. Stationarity for the $P$-objective and the face equality gives the displayed balance, whose nonnegative coverage forces $b \geq 1$. □

Thus trace extension supplies the balance identity used in Proposition 5.37. The unconditional Lemma 5.34 no longer requires trace extension or any choice of face dual; the stronger observed fact that every positive dual coalition has trace exactly $\{k\}$ is not required there either.

For the three-player branch the primal obstruction collapses to one scalar inequality. Put $R = N \setminus A = \{i, j\}$ and $d_A = v(N) - v(A)$. For $y \in \mathrm{Core}_A(v)$ and nonempty $D \subseteq R$, define the outside demand

$$\ell_D(y) = \max_{B \subseteq A} \{v(B \cup D) - y(B)\}.$$

**Proposition 5.39** (two-player-complement trace criterion)**.** *Let $v$ be exact, let $A = N \setminus \{i, j\}$, and let $y \in \mathrm{Core}_A(v)$. Then $y$ extends to a point of $F_A$ if and only if*

$$\ell_{\{i\}}(y) + \ell_{\{j\}}(y) \leq d_A. \tag{SD}$$

*In particular, exactness rules out every obstruction involving a coalition that contains both omitted players.*

*Proof.* An extension chooses two coordinates $z_i, z_j$ with $z_i + z_j = d_A$. Constraints whose traces on $R$ are $\{i\}$ or $\{j\}$ are exactly $z_i \geq \ell_{\{i\}}(y)$ and $z_j \geq \ell_{\{j\}}(y)$, so such coordinates exist exactly when (SD) holds. Constraints with empty trace already hold because $y \in \mathrm{Core}_A(v)$.

It remains to check a trace equal to $R$. For each $B \subseteq A$, exactness provides $a \in \mathrm{Core}(v)$ with $a(B) = v(B)$. Core feasibility and efficiency give

$$v(B \cup R) - v(B) \leq a(R) = v(N) - a(A) \leq v(N) - v(A) = d_A.$$

Since $y(B) \geq v(B)$, this implies $v(B \cup R) - y(B) \leq d_A$. Thus every two-player-trace constraint is automatic, completing the equivalence. □

Consequently equality in (TE) is equivalent to the restricted face LP having an optimum $y$ that satisfies (SD). This replaces a high-dimensional extension question by the exclusion of two witness coalitions, one carrying $i$ and one carrying $j$.

There is an exact dual for that last selection problem. Put

$$m_A = \min_{y \in \mathrm{Core}_A(v)} y_k, \qquad \Gamma_A = \min_{\substack{y \in \mathrm{Core}_A(v) \\ y_k = m_A}} \{\ell_{\{i\}}(y) + \ell_{\{j\}}(y) - d_A\}.$$

Then equality in (TE) is equivalent to $\Gamma_A \leq 0$.

**Lemma 5.40** (split-obstruction dual)**.** *The number $\Gamma_A$ is the maximum of*

$$\sum_C \tau_C v(C) + \sum_{p\in\{i,j\}} \sum_{B\subseteq A} \theta_{pB} v(B\cup\{p\}) - v(N) - (\alpha-1)v(A) - \beta m_A \tag{SDD}$$

*over $\tau_C, \theta_{pB} \geq 0$ and $\alpha, \beta \in \mathbb{R}$ satisfying*

$$\sum_B \theta_{pB} = 1 \quad (p = i, j)$$

*and the lifted incidence balance*

$$\sum_C \tau_C \mathbf{1}_C + \sum_{p\in\{i,j\}} \sum_{B\subseteq A} \theta_{pB} \mathbf{1}_{B\cup\{p\}} = \mathbf{1}_N + (\alpha-1)\mathbf{1}_A + \beta \mathbf{1}_{\{k\}}. \tag{SDB}$$

*Here $C$ ranges over the proper coalitions contained in $A$.*

*Proof.* Minimize $u_i + u_j - d_A$ subject to $y \in \mathrm{Core}_A(v)$, $y_k = m_A$, and

$$y(B) + u_p \geq v(B\cup\{p\}) \qquad (B \subseteq A,\ p = i, j).$$

At optimum $u_p = \ell_{\{p\}}(y)$. In the LP dual, stationarity of $u_p$ gives $\sum_B \theta_{pB} = 1$. Stationarity of the coordinates of $y$ first gives the corresponding balance on the traces $B$; replacing each trace by $B\cup\{p\}$ adds one unit on each omitted player and yields (SDB). The dual objective, after subtracting $d_A = v(N) - v(A)$, is (SDD). Strong duality completes the proof. □

The split dual can be compared with the simpler dual for $m_A$. Write that dual as

$$\sum_C \kappa_C \mathbf{1}_C = c\mathbf{1}_A + \mathbf{1}_{\{k\}}, \qquad \sum_C \kappa_C v(C) = cv(A) + m_A, \tag{MD}$$

with $\kappa_C \geq 0$ on proper coalitions contained in $A$.

The coefficient condition also has a one-dimensional primal form. Fix a minimum-coordinate dual coefficient $c$ and define the extended-real value

$$\begin{aligned}\widehat{\Gamma}_{A,c}(t) = \min\{u_i + u_j + t - d_A :\ & y(A) = v(A) + t,\ y_k = m_A - ct,\\ & y(B) \geq v(B) \quad (B \subsetneq A), \qquad y(B) + u_p \geq v(B\cup\{p\}) \quad (B \subseteq A,\ p = i, j)\}.\end{aligned} \tag{PT}$$

and put the value equal to $+\infty$ when the constraints are infeasible. Thus $\widehat{\Gamma}_{A,c}(0) = \Gamma_A$.

**Proposition 5.41** (minimum-dual tangent criterion)**.** *The optimal split-dual face contains a member satisfying $\alpha - 1 = \beta c$ if and only if*

$$\widehat{\Gamma}_{A,c}(t) \geq \Gamma_A \qquad \text{for every } t \in \mathbb{R}. \tag{TN}$$

*Equivalently, compatibility says that $t = 0$ globally minimizes the convex parametric split value in (PT).*

*Proof.* Write $\gamma = \alpha - 1$ and let $J$ be the split-dual objective at $t = 0$. The dual feasible region is independent of $t$, while its objective at $t$ is

$$J + t(\beta c - \gamma).$$

Hence $\widehat{\Gamma}_{A,c}$ is the supremum of affine functions and is convex. Its subdifferential at zero is the set of slopes $\beta c - \gamma$ attained on the optimal dual face: that face is convex and the slope map is affine, so its image is already an interval. Therefore $0 \in \partial\widehat{\Gamma}_{A,c}(0)$ exactly when some optimal dual has $\gamma = \beta c$. The convex optimality criterion identifies this with (TN). □

**Proposition 5.42** (compatible minimum-cover certificate)**.** *Suppose an optimal split dual in Lemma 5.40 and an optimal minimum-coordinate dual in* (MD) *satisfy*

$$\alpha - 1 = \beta c.$$

*If there is $z \in \mathrm{Core}(v)$ with $z(C) = v(C)$ whenever $\kappa_C > 0$, then $\Gamma_A \le 0$ and trace extension holds. When $v$ is exact and $c = 0$, the common-tightness condition is automatic.*

*Proof.* Let $\omega$ denote the aggregate coalition weights on the left side of (SDB). The coefficient identity and (MD) give the signed incidence identity

$$\sum_C \omega_C \mathbf{1}_C - \beta \sum_C \kappa_C \mathbf{1}_C = \mathbf{1}_N.$$

Because $\omega_C \ge 0$, core feasibility can only increase the $\omega$-sum when worths are replaced by $z$-values. Common tightness leaves the $\kappa$-sum unchanged, irrespective of the sign of $\beta$. Hence

$$\sum_C \omega_C v(C) - \beta \sum_C \kappa_C v(C) \le \sum_C \omega_C z(C) - \beta \sum_C \kappa_C z(C) = v(N).$$

Using (MD) and $\alpha - 1 = \beta c$, the left side minus $v(N)$ is exactly the optimal objective (SDD), namely $\Gamma_A$. Thus $\Gamma_A \le 0$.

If $c = 0$, the nonnegative incidence identity in (MD) forces $\kappa_{\{k\}} = 1$ and every other $\kappa_C = 0$. Exactness at $\{k\}$ then supplies the required $z$. □

The other hypothesis also has a scalar primal form. For a fixed optimal minimum cover $\kappa$, its positive support is jointly tight at a full-core point if and only if

$$\min_{z \in \mathrm{Core}(v)} \{cz(A) + z_k\} = cv(A) + m_A. \tag{NL}$$

Indeed, (MD) makes the left side at least the right side, and equality is equivalent to zero total $\kappa$-weighted core slack. Thus the remaining compatible-cover route is entirely primal: prove tangent non-descent (TN) and the minimum-normal lift (NL) from star failure and the leximin strata. These two statements concern the same face. Every feasible $y$ in (PT) satisfies

$$\sum_C \kappa_C [y(C) - v(C)] = c[y(A) - v(A)] + (y_k - m_A) = ct - ct = 0. \tag{MF}$$

All summands are nonnegative, so every positive $\kappa$-coalition is tight along the entire tangent LP. Hence tangent non-descent is an optimization statement on the restricted minimum-cover face, while (NL) says that this same tight face meets the full core.

This also gives an entirely primal proof of Proposition 5.42. Let $z$ attain (NL) and put $t = z(A) - v(A)$. Equality in (NL) gives $z_k = m_A - ct$, so the restriction of $z$ to $A$, together with $u_i = z_i$ and $u_j = z_j$, is feasible in (PT). Full efficiency gives $z_i + z_j = d_A - t$, hence this feasible point has objective zero. Therefore

$$\Gamma_A = \widehat{\Gamma}_{A,c}(0) \le \widehat{\Gamma}_{A,c}(t) \le 0$$

by tangent non-descent.

Exactness alone does not force the normal lift, even at unit price. Define

$$\Delta^{\mathrm{NL}}_{A,k} = \min_{z\in \mathrm{Core}(v)} \{z(A) + z_k\} - [v(A) + m_A] \qquad (c = 1).$$

The most natural quantitative comparison would be

$$\Delta^{\mathrm{NL}}_{A,k} \le \eta_S(\phi). \tag{SN}$$

It is false, and its sharp-bound deformation also refutes star-conditioned trace extension itself.

**Proposition 5.43** (exact star–trace counterexample)**.** *For $0 \le \varepsilon \le 1$, put*

$$\begin{aligned} p &= (13, 13, -12, -3, 13, 0, 1),\\ q^{\varepsilon} &= (11, 14 + \varepsilon, -3, 2 - \varepsilon, 16, -10, -5),\\ u &= (19, 19, -20, -3, 20, 2, -12), \end{aligned}$$

*and let $v^{\varepsilon}$ be their lower-envelope game, with common grand worth 25. Let $S = \{2, 4, 5\}$, $k = 5$, $A = \{1, 3, 5, 6, 7\}$, and $P = A \setminus \{5\}$.*

*At $\varepsilon = 1/10$, the restricted minimum has a unit-price dual but*

$$\Delta^{\mathrm{NL}}_{A,5} = 1 > \frac{17}{20} = \eta_S(\phi),$$

*so (SN) fails. At $\varepsilon = 2/3$, the star condition fails at the leximin point, $S$ is binding, and nevertheless*

$$\max_{y\in \mathrm{Core}_A(v)} y(P) = -5 > -6 = \max_{x\in F_A} x(P). \tag{CX}$$

*Thus trace extension fails, with trace gap 1 and split obstruction $\Gamma_A = 2$.*

*Proof.* All three generators have sum 25 and belong to the core of their lower-envelope game. Every proper-coalition worth is attained by a generator, so $v^{\varepsilon}$ is exact.

At $\varepsilon = 1/10$, exact primal–dual LP pairs give restricted minimum 14, the unit cover $\kappa_{\{3,5,6\}} = \kappa_{\{1,5,7\}} = 1$, normal-lift gap 1, and star opening 17/20.

At $\varepsilon = 2/3$, the leximin point is

$$\phi = \left(\frac{43}{3}, 16, -\frac{25}{2}, 0, \frac{52}{3}, -\frac{5}{2}, -\frac{23}{3}\right).$$

Its successive gain levels are $3, 10/3, 13/3, 15/2$, pinning respectively $\{2, 4\}, \{1\}, \{5, 7\}, \{3, 6\}$; exact primal–dual pairs certify each leximin stage. The core point $u$ has $u(S) = 36 = \phi(S) + 8/3$, so $S$ is binding. For the star LP, unit weights on coalitions $\{1, 2, 3, 6, 7\}$ and $A$, unit weight on the floor of player 4, and grand coefficient $-2$ give an exact dual upper bound $x(S) \le 100/3 = \phi(S)$. Hence the star opening is zero; the corresponding residual vector on $S$ is $(1, 2, 1)$.

For trace extension, the restricted witness $(y_1, y_3, y_5, y_6, y_7) = (13, -5, 14, -8, -5)$ attains $y(P) = -5$; unit dual weights on $\{3, 5, 6\}$ and $\{1, 5, 7\}$ certify optimality. On the full face, the witness

$$(13, 47/3, -38/3, 1/3, 15, -4/3, -5)$$

attains $x(P) = -6$. Half-unit dual weights on $\{3, 5, 6\}, \{1, 3, 5, 6\}, \{2, 5, 7\}, \{1, 4, 5, 7\}$, together with grand and face coefficients $-1/2$ and $-3/2$, certify that this is the full-face maximum. This proves (CX); the exact split LP gives $\Gamma_A = 2$. □

The counterexample was missed by broad but non-adaptive screens. Across 10,395 arbitrary codimension-two faces of 99 nondegenerate exact seven-player lower-envelope games, 86 restricted optima did not extend, with exactly certified gaps of 3 and 6 in the two control corpora. Yet none of those gaps occurs among the 501 faces conditioned on star failure. The largest exact ratio of trace gap to star opening in the first control is 5, so the tempting quantitative bound "gap at most twice the opening" was already false. Six rational affine bridges from a genuine three-level failure to an exact trace-gap control tell the same story: every zero-opening grid point has zero gap, while the first positive gap appears only at $t = 17/20$ or $9/10$, after star opening is already positive. No gap occurs in the six exact three-player failure cases or in five additional float multilevel candidates from 1,484 nondegenerate targeted games; exactification sent the sampled extremals from the latter set to the two-level branch. A further 1,000-game local mutation screen retained 54 genuine three-level failures with a unique minimum residual. All 54 have zero trace gap and positive face-target margin; two rational extremals certify target margins $15/4$ and 2.

The decisive search was instead transverse to the exact positive equality $\Delta^{\mathrm{NL}}_{A,5} = \eta_S(\phi) = 1$. Of 252 two-scale, sum-preserving generator deformations, all retained a returned unit-price cover and positive normal gap, while 88 violated (SN) in floating arithmetic. The ten closest records all recomputed to exact violations. On the rational path in Proposition 5.43, the normal gap stays 1 and $\Gamma_A$ stays 2; the exact star opening is $1/40$ at $\varepsilon = 13/20$ and reaches zero at $\varepsilon = 2/3$. Therefore star failure alone does not imply trace extension, and this unqualified trace route is closed. The counterexample has residuals $(1, 2, 1)$, however, so it belongs to the two-level branch already settled by Proposition 5.3. It does not refute the unresolved genuinely multilevel program. Proposition 5.32 then suggests the sharp repair.

**Conjecture 5.44** (multilevel star-conditioned trace extension)**.** *Let $v$ be exact, let $|S| = 3$, suppose $(\star_S)$ fails at the leximin point, and suppose no failure cover has at most two residual levels on $S$. Let $k$ be the unique minimum-residual player supplied by Proposition 5.32, put $A = (N \setminus S) \cup \{k\}$ and $P = A \setminus \{k\}$. Then equality holds in (TE).*

This is now a seven-case exact frontier. An exact whole-face audit now verifies the quantifier in the conjecture, rather than merely inspecting one returned cover: for each of the three residual pairs, adding the equality $r_i = r_j$ to the rational failure-cover polyhedron is infeasible in all seven cases. Hence every cover has three distinct levels. Their strict residual orders are fixed, and after normalizing the minimum residual to one, the exact minimum gaps to the other two coordinates are one and two. The original six cases have $\Gamma_A = 0, 0, 0, -3, 0, 0$, the seventh case in Proposition 5.35 has zero trace gap, and the 54 genuine three-level local mutations all have zero trace gap. Thus the seventh case falsifies the stronger face target while continuing to satisfy this trace-extension conjecture. The earlier positive-trace-gap counterexample passes the control check: one of its pair-tie systems is feasible, so it is genuinely outside the conjectured branch.

The new counterexample shows that the multilevel hypothesis cannot simply be deleted. A targeted tie-splitting audit makes 378 exact sum-preserving deformations around the counterexample. Of these, 246 remain binding star failures; every recovered failure cover still has exactly two residual levels, and none enters the three-level chamber. A second transverse screen makes 756 deformations around two distinct whole-face multilevel fixtures. It retains 181 binding star failures and 164 returned three-level unique-minimum cases, with no positive trace gap; the extremum from each fixture recomputes exactly to gap zero and remains whole-face genuinely multilevel. Thus the positive trace gap appears locally confined to the already-settled boundary.

A separate 3,528-deformation screen targeted the high-coordinate sign rather than trace gap. It retained 1,275 binding star failures and 1,146 returned three-level failures. Two had negative float high-coordinate margins; the worst recomputed exactly to the $-2/3$ face and coordinate margins in Proposition 5.35. This closes the stronger face-target route but leaves both the minimum-residual carrier inequality and multilevel trace extension intact in the new case.

The new criterion identifies the missing step more sharply. The exact $\Gamma_A$ values in the six three-level failures are $0, 0, 0, -3, 0, 0$. By contrast, the exactly certified worst-gap points in the two controls have $\Gamma_A = 3$ and 6, while the separate ratio-5 control has $\Gamma_A = 5$ even though the first returned restricted optimum has excess 6. All have zero two-player-trace excess, as Proposition 5.39 requires. The counterexample shows that star failure need not force $\Gamma_A \le 0$ without the multilevel hypothesis in Conjecture 5.44.

A tempting point-free strengthening is false even inside the hard corpus. Maximize each marginal $v(B \cup \{p\}) - v(B)$ separately over traces $B$ that contain $k$ and no other member of $S$. In one exact three-level star failure the two maxima exceed $d_A$ by 3, although $\Gamma_A = 0$: the selected restricted optimum places exactly compensating slack on one offending trace. Thus a universal marginal-worth inequality cannot prove trace extension; selection inside the restricted optimal face, equivalently the dual balance (SDB), is essential.

The compatible-cover certificate gives proof-producing decompositions for all six exact three-level cases. Every optimal split-dual face contains a member with $\alpha - 1 = \beta c$, and every minimum-cover support has exact common-tightness gap zero. Three cases have $c = 0$ and hence follow from singleton exactness directly; the three $c = 1$ cases have exact rational common core witnesses. In the sixth case the dual first returned by the LP does not have the coefficient match, but an exact search of its optimal-dual face finds the compatible representative $\alpha - 1 = \beta = 1$.

No optimal-dual face of a positive-gap control contains a compatible member. The worst-gap controls also have common-tightness gap 2; the ratio-5 control has a common-tight minimum support but fails coefficient compatibility throughout its optimal split-dual face. Primal tangent minimization gives the same separation: the base parameter minimizes (PT) in all six hard cases, whereas the three distinct positive-gap controls have exact tangent improvements 1, 6, and 4, attained at $t = 1$, 11, and 4. The stronger demand that the split weights dominate $\beta\kappa$ coefficientwise is feasible in only one of the six hard cases, so the signed common-tightness argument is genuinely needed. The counterexample has $c = 1$, normal-lift gap 1, no compatible optimal split dual, and exact tangent improvement 2. Hence neither tangent non-descent nor the minimum-normal lift follows from star failure and the leximin strata alone. Conjecture 5.44 excludes the counterexample's two-level residual pattern; proving or breaking that repaired statement is now the precise face-lifting target.

The cover-independent audit reveals a substantially simpler target. It uses the full minimum-residual class at once.

**Conjecture 5.45** (minimum-residual-class carrier)**.** *Let $v$ be exact and let $(\star_S)$ fail at $\phi$. If the failure is not already covered by Proposition 5.3, then there is a failure cover with minimum-residual class $K = \{i \in S : r_i = r_0\}$ and $P = N \setminus S$ such that*

$$[\phi(P \cup K) - v(P \cup K)] \leq [\phi(K) - v(K)], \tag{KC}$$

*or equivalently*

$$v(P \cup K) - v(K) \geq \phi(P).$$

This single inequality would close the multilevel capacity branch immediately: apply Proposition 5.5 with $C = K$. It requires no singleton selector and is insensitive to how payoffs are ranked inside $K$. A deduplicated exact audit verifies (KC) for all 669 stored multilevel game–coalition pairs (702 records): 644 inequalities are strict and 25 are equalities, with margins ranging from $-28$ to 0. This includes all 505 adversarial mutation pairs on which the fixed-cover escape test mostly fails. Thus (KC), rather than a singleton rule or residual-gap escape, is now the sharp computational theorem target. Deterministic replay of the historical random streams recovered and verified the discarded point generators for all 25 equality records. They represent 23 games, and every one is the lower envelope of exactly two *distinct* common-sum points; the nominal three-generator records merely repeat one point. Hence every non-strict case in this exact corpus already lies in the model class settled by Proposition 5.28. Outside that proved boundary subcase, all audited class certificates are strict.

Lemma 5.29 gives an exact-game face reduction; the finite-model audit is one way to expose its witnesses. Among the 530 deduplicated class-carrier records that retain their point generators, 461 are direct: some generator attaining $v(P \cup K)$ already has $a(P) \geq \phi(P)$. Each of the remaining 69 records has exactly three distinct generators, a unique $P \cup K$ minimizer, $|K| = 3$, and a strictly positive trace-compensated margin between $19/4$ and 8. All 69 arise from one seeded target family. These counts are exact finite evidence, not a three-generator theorem, but they isolate the only nontrivial generator mechanism in the embedded corpus.

The quantifier over failure covers in Conjecture 5.45 is essential. The next exact three-generator family has failure covers in three residual chambers. On an open parameter interval, two chambers select a class that violates (KC), while the third selects the unique carrier class. Thus no distinguished output of a failure-cover LP can be used without a separate selection argument.

**Proposition 5.46** (failure-cover selection is essential)**.** *For $0 \le \kappa \le 1$, let $v^\kappa$ be the lower envelope of*

$$\begin{aligned} p^\kappa &= (-4, 3-\kappa, 27, \kappa, 10, -8), \\ q^\kappa &= (-4, 1-\kappa, 29, -2+\kappa, 12, -8), \qquad v^\kappa(T) = \min\{p^\kappa(T), q^\kappa(T), u^\kappa(T)\}. \\ u^\kappa &= (-\kappa, -1, 22, \kappa, 10, -3), \end{aligned}$$

*Then $v^\kappa$ is exact and*

$$\phi^\kappa = \left(-\frac{5}{2} - \frac{\kappa}{2}, \frac{1}{2} - \frac{\kappa}{2}, \frac{51}{2} - \frac{\kappa}{2}, -1+\kappa, 11, -\frac{11}{2} + \frac{\kappa}{2}\right).$$

*For $S = \{2,3,4,6\}$ and $P = \{1,5\}$, the coalition $S$ is binding by $5/2 - \kappa/2$, but $(\star_S)$ fails throughout the interval. Moreover every normalized failure cover has, for some*

$$a \ge 1, \qquad t - a \ge 1,$$

*the residual vector*

$$(r_2, r_3, r_4, r_6) = (a, t, t-a, a+t), \tag{RC}$$

*and every vector in this cone is attained by a failure cover. Its minimum-residual class is*

$$K = \begin{cases} \{2\}, & t > 2a, \\ \{2,4\}, & t = 2a, \\ \{4\}, & a+1 \le t < 2a. \end{cases}$$

*Writing*

$$\Delta_\kappa(K) = [\phi^\kappa(P \cup K) - v^\kappa(P \cup K)] - [\phi^\kappa(K) - v^\kappa(K)],$$

*the three possible margins are*

$$\Delta_\kappa(\{2\}) = -\frac{3}{2} + \frac{\kappa}{2}, \qquad \Delta_\kappa(\{4\}) = \Delta_\kappa(\{2,4\}) = \frac{1}{2} - \frac{\kappa}{2}.$$

*Thus, for $0 \le \kappa < 1$, the normalized cover with residuals $(1,2,1,3)$ selects the noncarrier class $\{2,4\}$, whereas the cover with residuals $(1,3,2,4)$ selects the carrier class $\{2\}$. At $\kappa = 1$ the former class reaches carrier equality while the latter remains strict.*

*Proof.* All three generators have total 28. Each is therefore efficient and dominates $v^\kappa$ on every coalition, hence belongs to the core; a generator attains $v^\kappa(T)$ for every $T$, proving exactness. The singleton floors are

$$m^\kappa = (-4, -1, 22, -2+\kappa, 10, -8).$$

We verify the leximin point directly. If every gain $x_i - m_i^\kappa$ is at least $s$, the core inequality $x(\{1,2,3,6\}) \ge 18 - \kappa$ and efficiency imply $x_4 + x_5 \le 10 + \kappa$, while the floors imply $x_4 + x_5 \ge 8 + \kappa + 2s$. Thus $s \le 1$, and equality pins $(x_4, x_5) = (-1+\kappa, 11)$. On that face, $x(\{3,4,6\}) \ge 19 + \kappa$ gives $x_1 + x_2 \le -2 - \kappa$; hence the next gain is at most $(3-\kappa)/2$, with equality pinning $(x_1, x_2) = (-5/2 - \kappa/2, 1/2 - \kappa/2)$. On the resulting face, $x(\{1,3,4\}) \ge 22$ gives $x_6 \le -11/2 + \kappa/2$, so the next gain is at most $5/2 + \kappa/2$. Equality pins $x_6$, and efficiency fixes $x_3 = 51/2 - \kappa/2$. Direct affine endpoint substitution shows that the displayed point is in the core throughout $0 \le \kappa \le 1$, so these successive bounds prove the formula for $\phi^\kappa$.

The proper coalitions tight at $\phi^\kappa$ are exactly

$$134, \quad 125, \quad 45, \quad 1236, \quad 346,$$

where strings denote player sets. For each other proper coalition, one fixed generator lies strictly below $\phi^\kappa$ at both parameter endpoints, so the same affine strict inequality holds throughout the interval. The exact verifier records all 62 selectors. The multipliers

$$\lambda_{134} = 1, \qquad \lambda_{125} = 2, \qquad \lambda_{45} = 1, \qquad \mu = 3$$

cover the two players in $P$ exactly $\mu$ times and leave residuals $(1, 2, 1, 3)$ on $S$. If $d$ were a direction in $(\star_S)$, then

$$r \cdot d_S = \mu d(N) - \sum_T \lambda_T d(T) \le 0,$$

but $d_i \ge 0$ on $S$ and $d(S) > 0$ make the left side positive. Hence $(\star_S)$ fails. Also $v^\kappa(P) = 6$, so every core point has $x(S) = 28 - x(P) \le 22$; the generator $p^\kappa$ attains equality, while $\phi^\kappa(S) = 39/2 + \kappa/2$. This proves the binding margin $5/2 - \kappa/2$.

It remains to describe the whole cover cone. Put multipliers $A, B, C, D, E$ on the five tight coalitions in the displayed order. Exact coverage of players 1 and 5 gives

$$\mu = A + B + D = B + C, \qquad \text{so} \qquad C = A + D.$$

Setting $a = A$ and $t = B - E$ now gives

$$(r_2, r_3, r_4, r_6) = (a, t, t - a, a + t).$$

The normalization $r_i \ge 1$ reduces to $a \ge 1$ and $t - a \ge 1$. Conversely, every such pair is realized by taking $D = E = 0$, $A = C = a$, and $B = t$. Comparing the four coordinates yields the three stated residual chambers. Direct substitution in the worth function gives the displayed affine values of $\Delta_\kappa(K)$. Finally, $(a, t) = (1, 2)$ and $(1, 3)$ give the two explicit covers claimed. □

A dedicated screen of 500 six-player, three-generator nearby instances retained 496 games with three distinct generators. It found 30 multilevel binding star failures and one positive margin for the cover returned by the search LP; exactification produced the $\kappa = 0$ member of Proposition 5.46. An exact prescribed-class audit then tested every proper class and found the favorable $K = \{2\}$ cover above. Two targeted mutation sweeps around that member retained 1,998 nondegenerate games and found 28 multilevel binding star failures. The returned cover failed in 16, but full chamber enumeration found a carrier cover in every case; the closest exact mutation is the $\kappa = 1$ endpoint. These screens are finite discovery evidence; the proposition itself is analytic and exhaustive over the full cover cone for every parameter value.

A separate targeted screen of 3,000 four-generator lower-envelope games found 1,123 star failures and 25 genuinely multilevel failures, with no positive class-carrier margin; the closest boundary case was recomputed exactly and has margin zero. The broader screen is discovery evidence only, whereas the 669-pair audit is fully exact. The boundary admits an exact parametric analysis.

**Proposition 5.47** (class-carrier boundary coupling)**.** *For $0 \le \varepsilon \le 1/3$, let $v^\varepsilon$ be the lower envelope of the four common-sum points*

$$p = (1, 1, 1, 2, 1, 2), \quad q^\varepsilon = (0, 1, 2 - \varepsilon, 0, 4 + \varepsilon, 1),$$
$$r = (2, 1, 2, 1, 2, 0), \quad s = (1, 2, 0, 2, 1, 2).$$

*Put* $S = \{2, 4, 5, 6\}$, $P = \{1, 3\}$, *and* $K = \{4, 6\}$. *Then* $v^\varepsilon$ *is exact,*

$$\phi(v^\varepsilon) = (1, 2, 1, 1, 2, 1),$$

*the class-carrier failure margin is exactly*

$$[\phi(P \cup K) - v^\varepsilon(P \cup K)] - [\phi(K) - v^\varepsilon(K)] = \varepsilon,$$

*and the star opening is exactly*

$$\eta_S(\phi) = 3\varepsilon.$$

*Thus crossing the class-carrier boundary in this exact family restores* $(\star_S)$ *at three times the crossing rate.*

*Proof.* The generators have common total 8, so their lower envelope is exact. The singleton floors are $(0, 1, 0, 0, 1, 0)$. Direct endpoint substitution shows that $\phi = (1, 2, 1, 1, 2, 1)$ belongs to the core throughout the interval. Its six gains all equal one, while the total gain is six; hence it is the leximin point. Moreover,

$$v^\varepsilon(K) = 1, \qquad v^\varepsilon(P \cup K) = 3 - \varepsilon, \qquad \phi(P) = 2,$$

which proves the carrier-margin formula.

The allocation

$$y^\varepsilon = \phi + \varepsilon(-1, 0, -2, 1, 0, 2)$$

belongs to the core for $0 \le \varepsilon \le 1/3$, dominates $\phi$ on $S$, and satisfies $y^\varepsilon(S) = 6 + 3\varepsilon$. Core membership is a finite affine endpoint check: for each proper coalition the same one of the four generators lies weakly below $y^\varepsilon$ at both endpoints, hence throughout the interval. The standalone exact verifier records all 62 selectors.

For the reverse inequality, let $x$ be any core point with $x_i \ge \phi_i$ for $i \in S$. The three core constraints and two coordinate floors

$$x_{34} \ge 2 - \varepsilon, \quad x_{145} \ge 4, \quad x_{136} \ge 3 - \varepsilon, \quad x_2 \ge 2, \quad x_5 \ge 2$$

give, after taking multiplicities $(1, 1, 2, 2, 1)$,

$$3x(P) + 2x(S) = 24 - x(S) \ge 18 - 3\varepsilon.$$

Thus $x(S) \le 6 + 3\varepsilon$, and $y^\varepsilon$ attains equality. □

The recovered two-generator boundary has a second exact local test. Six of the 25 equality records had stored a duplicated generator. A 1,620-direction single-copy splitting screen found positive minimum-class margin in exactly those six records. In the worst screened direction for each record, exact constraint-generation certificates give class margin $1/100$ and star opening $1/50$, hence the same ratio 2 in all six cases. These are class-specific comparisons: another carrier has margin zero in every deformation, so the global $\gamma_S$ defined above remains zero. The calculation therefore supplies further boundary coupling for (KC), not an improved coefficient in (Q). The exact leximin verifier begins with singleton and co-singleton core constraints and adds every violated omitted coalition until its rational optimizer satisfies the full core.

Proposition 5.3 explains an important part of this structure exactly. If the residuals have only the levels 1 and $R > 1$, the entire minimum-residual class $C = \{i \in S : r_i = 1\}$ is itself

the required face certificate; no crossing-cover construction is needed. The rare same-stratum pair exceptions to singleton certification are examples with $|C| = 2$. Accordingly, all unresolved certificate geometry lies in duals with at least three distinct residual levels. Proposition 5.5 removes a second large class: it is enough that an off-side carrier have no more slack than its trace inside $S$; tightness of the carrier is a particularly simple sufficient condition. In particular, the three-level witness `WITNESS_KEY_FALSE_n6` is certified by $T = \{1, 2, 6\}$ and $C = \{1\}$ (player labels one-indexed), even though the union-of-cover certificate described above fails.

Proposition 5.2 removes the frequency language from the logical frontier. Every pair $(v, S)$ falls into the exact dichotomy

$$d_v^*(S) \le \sigma_S \quad \text{or} \quad d_v^*(S) > \sigma_S.$$

The first side admits no binding exact bump and is finished by Lemma 5.31. On the second side a binding exact bump exists, and the missing structural statement is precisely

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Proposition 5.3 proves this implication whenever the failure dual has at most two residual levels, and Corollary 5.4 covers every $|S| = 2$. The remaining capacity frontier therefore requires both $|S| \ge 3$ and a genuinely three-or-more-level dual. After Proposition 5.5, a counterexample to the capacity implication must additionally have no off-side slack-dominating carrier. The singleton/pair certificate statistics motivate how to attack those cases, but neither their observed percentages nor sampling rarity enters the conditional theorem.

The sharpened computational target is correspondingly narrow. The broad slack-domination condition covered all 141 saved hard star-failure cases at $n = 6, 7$ and all 744 fresh binding star failures in a screen of exact lower-envelope games at $n = 4, 5, 6$, with no exception. The new class-carrier audit finds a successful full minimum-residual class in all 669 distinct exact pairs. Proposition 5.46 shows that this class need not come from an arbitrarily returned failure cover. This remains evidence, not a theorem, but proving Conjecture 5.45 would close that branch with a single application of Proposition 5.5. Proposition 5.28 proves the broader existence of some slack-dominating carrier for every two-generator lower-envelope game. Thus within the finite lower-envelope model class, any counterexample must use at least three genuinely active generators; this is a structural restriction, not a sampling observation.

The obvious three-generator escape from the sharp two-generator boundary is itself blocked, but for an instructive reason. In the exact equality fixture, let $S = \{1, 2, 3, 4, 6\}$ and transfer $a = 1/10$ from the off-side player 5 of the second generator equally among the five members of $S$, retaining both original generators. All three generators are genuinely active. The leximin point remains $(1, 4, 3, 5, 7, 4)$ and the minimum off-side slack comparison becomes $1/50 > 0$, so the carrier condition has indeed been crossed. However, $(\star_S)$ is simultaneously restored: the core point

$$(26/25, 4, 153/50, 253/50, 171/25, 4)$$

dominates $\phi$ on $S$ and has $S$-sum $429/25 > 17$. Thus the star opening is $4/25$, eight times the carrier escape. This is an exact boundary calculation, not a general theorem; a finite simplex-grid deformation screen found the same coupling throughout the tested local family. It suggests proving the carrier conjecture contrapositively: show that escaping every off-side carrier inequality constructs a core direction nonnegative on $S$, rather than trying to track one distinguished minimizing generator.

The carrier lift suggests a sharp quantitative form of that contrapositive:

$$\eta_S(\phi) \geq \gamma_S(\phi). \tag{Q}$$

Proposition 5.23 disproves (Q) in unrestricted dimension, while Proposition 5.22 proves it through six players. In view of Proposition 5.12, this sharp finite-dimensional theorem applies to every $\psi \in \mathrm{Core}(v)$, not only to $\psi = \phi$. Proposition 5.16 proves it for $|S| = 2$, and the coefficient one is already sharp there. Proposition 5.15 proves the stronger uniform lift when $|N \setminus S| = 1$; hence Corollary 5.17 settles (Q) for every exact game with at most four players, and Proposition 5.18 closes the remaining three-versus-two case by exact dual enumeration, even under the weaker assumption of total balancedness. Finally, Proposition 5.22 closes both new six-player splits by complete minimal-balanced enumeration. The three-generator lower envelope in Proposition 5.23 then violates (Q) at seven players at its leximin point with $(\gamma, \eta) = (3/4, 0)$ (and at the core reference zero with $(1, 0)$), making the six-player theorem sharp. Earlier, six exact lower-envelope screens covering 3,650 games and 35,629 binding pairs $(v, S)$, all 746 cases with $\gamma_S(\phi) > 0$ satisfied (Q); the minimum in every screen was recomputed exactly and equalled one. These counts are now historical cross-checks, not part of the proof. The exactness hypothesis is load-bearing even for the coefficient: a balanced non-exact five-player game has leximin point $(-3/2, 10, 6, 17/2, -10)$ and, at $S = \{2, 3, 4\}$, $\gamma_S = 7/2$ but $\eta_S = 1/2$. Thus the dimension-free target is a convex-combination statement for the carrier-tight points supplied by Lemma 5.11, not an inequality valid on arbitrary balanced cores.

Earlier independent screens anticipated the theorem and tested whether it depended on the leximin reference. At the leximin point, 10,000 four-player games yielded 4,473 positive-$\gamma$ three-player targets, including exact equality, and 3,000 five-player games yielded 3,362 positive-$\gamma$ targets of size at least three; none had $\eta/\gamma < 1$, and the exact five-player minimum was 91/83. More strikingly, the leximin hypothesis was then discarded: 10,000 exact four-player core references yielded 19,521 positive-$\gamma$ cases with exact minimum one, while 6,400 exact five-player references yielded 3,732 positive-$\gamma$ targets of size at least three, again with no violation. Finally, 800 exact references drawn from four saved five-player games with genuinely non-large cores produced 729 positive-$\gamma$ targets of size at least three, no violation, and exact minimum ratio one. Thus the pattern is not explained merely by the implication from exactness to largeness that holds in smaller player sets. These finite campaigns do not add logical scope to Propositions 5.18 and 5.22, but they helped identify the selection-free exact-core closure inequality (CG)–(CE).

The common-coordinate strengthening $\kappa_S(\psi) \geq \gamma_S(\psi)/|S|$ also survived dense, sparse, generator, and non-large exact screens in the first open regime $|S| = 3$, $|N \setminus S| = 2$. The largest generator campaign contained 60,418 positive-$\gamma$ targets and had exact minimum normalized ratio 47/45; the non-large audit attained equality. By contrast, a targeted 26,471-case screen drove the corresponding outside-slack ratio down to 3/112 while retaining carrier-normalized ratio 3/2. Thus a proof cannot replace the carrier slack system by a single outside-coalition cap; the exact dual in Proposition 5.13 is the sharper remaining object.

At six players the totally balanced analogue already fails, with $(\gamma, \eta) = (1, 0)$, but exact closure removes the gap. The resulting three-block identity, Lemma 5.20, first extends to the complete unit-triple layer of Proposition 5.21. More decisively, Lemma 5.19 converts every relevant eta-dual vertex into a minimal balanced collection containing $S$. Exhausting all 1,489 symmetry orbits and certifying every exact-cone optimum proves the quantitative carrier inequality for every exact game through six players (Proposition 5.22). The earlier direct sample of 505 orbits is recovered in full by the census and now serves as an independent cross-check

rather than merely finite evidence. The three-generator exact game in Proposition 5.23 has $(\gamma, \eta) = (3/4, 0)$ at its leximin point, so this finite theorem is dimension-sharp.

**Remark 5.48.** *Lemma 5.30 is a leximin parametric-monotonicity statement; its cousin the nucleolus is* not *coalitionally monotone, so exactness (Lemma 5.1) is doing real work and the "self-defeating within-S transfer" heuristic is not by itself a proof. For proper coalitions of size at least three we leave (D), (E′) open and record the conditional theorem:* on exact games, $\phi$ is monotone under proper nonsingleton bumps provided Lemma 5.30 holds*; singleton and grand-coalition bumps remain outside this reduction. The singleton case is settled positively by Theorem 2.2, while the grand case is settled negatively by Theorem 3.2; the necessity direction (Theorem 4.1) is also unconditional.*

## References


Dutta B, Ray D (1989) A concept of egalitarianism under participation constraints. Econometrica 57:615–635. https://doi.org/10.2307/1911055

Dietzenbacher B (2021) Monotonicity and egalitarianism. Games Econ Behav 127:194–205. https://doi.org/10.1016/j.geb.2021.03.006

Schmeidler D (1972) Cores of exact games, I. J Math Anal Appl 40:214–225. https://doi.org/10.1016/0022-247X(72)90045-5

Laplace Mermoud D, Grabisch M, Sudhölter P (2025) Minimal balanced collections and their applications to core stability and other topics of game theory. arXiv:2507.05898. https://arxiv.org/abs/2507.05898

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Math Methods Oper Res 95:35–80. https://doi.org/10.1007/s00186-022-00770-4